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AI-Powered Sperm Analysis from Microscopic Videos

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Abstract

This thesis explores the application of artificial intelligence to automate and enhance semen analysis through advanced computer vision and deep learning techniques. The system leverages YOLOv11 for real-time sperm detection and tracking in microscopic video sequences, offering high accuracy and speed necessary for dynamic environments. For species-level morphological classification of sperm cells, EfficientNet was employed due to its state-of-the-art performance in image-based classification tasks and its computational efficiency. A strong emphasis was placed on data preprocessing, including frame selection, noise reduction, contrast enhancement, and normalization, to ensure robust model performance across diverse image qualities. The study addresses several AI challenges in reproductive science, such as generalization across species, video quality inconsistencies, and the variability of biological samples. Ethical considerations surrounding the use of AI in biological and veterinary contexts are also examined. The results demonstrate that integrating deep learning into semen analysis can significantly improve the accuracy, consistency, and scalability of traditional methods. This work lays a foundation for future developments toward explainable, cross-species, and privacy-aware AI systems in reproductive diagnostics.

Keywords: Artificial Intelligence, Sperm Detection, YOLOv11, EfficientNet, Semen Analysis, Deep Learning

Résumé

Ce mémoire explore l'application de l'intelligence artificielle pour automatiser et améliorer l'analyse du sperme à l'aide de techniques avancées de vision par ordinateur et d'apprentissage profond. Le système utilise **YOLOv11** pour la détection et le suivi des spermatozoïdes en temps réel dans des séquences vidéo microscopiques, offrant une grande précision et la rapidité requise dans des environnements dynamiques. Pour la classification morphologique des spermatozoïdes par espèce, **EfficientNet** a été utilisé en raison de ses performances de pointe dans les tâches de classification d'images et de son efficacité computationnelle. Une attention particulière a été portée au prétraitement des données, incluant la sélection d’images, la réduction du bruit, l’amélioration du contraste et la normalisation, afin d’assurer des performances robustes du modèle face à la diversité des qualités d’image.

L’étude traite plusieurs défis liés à l’IA dans le domaine de la reproduction, notamment la généralisation entre espèces, les variations de qualité vidéo et la variabilité des échantillons biologiques. Les considérations éthiques liées à l'usage de l'IA dans les contextes biologiques et vétérinaires sont également abordées.

Les résultats montrent que l'intégration de l'apprentissage profond dans l'analyse du sperme peut considérablement améliorer la précision, la cohérence et la capacité de mise à l'échelle des méthodes traditionnelles. Ce travail constitue une base pour le développement futur de systèmes d'IA explicables, inter-espèces et respectueux de la vie privée dans le domaine du diagnostic reproductif.

Mots-clés : Intelligence Artificielle, Détection de Spermatozoïdes, YOLOv11, EfficientNet, Analyse du Sperme, Apprentissage Profond

الملخص

تتناول هذه الأطروحة استخدام تقنيات الذكاء الاصطناعي لأتمتة وتحسين تحليل السائل المنوي من خلال الرؤية الحاسوبية المتقدمة وتقنيات التعلم العميق. يعتمد النظام على نموذج **YOLOv11** للكشف عن الحيوانات المنوية وتتبعها في مقاطع الفيديو المجهرية، مع تحقيق دقة عالية وسرعة مناسبة للبيئات الديناميكية.ولتصنيف الخلايا المنوية حسب النوع بناءً على شكلها المورفولوجي، تم استخدام نموذج **EfficientNet** نظراً لأدائه المتفوق في مهام تصنيف الصور وكفاءته من حيث استهلاك الموارد.

تم إيلاء اهتمام خاص لمرحلة معالجة البيانات المسبقة، بما في ذلك اختيار الإطارات المناسبة، وتقليل التشويش، وتعزيز التباين، وتوحيد البيانات، وذلك لضمان أداء قوي للنموذج على أنواع مختلفة من جودة الصور.

تناقش هذه الدراسة عدداً من التحديات المرتبطة بتطبيق الذكاء الاصطناعي في مجال علوم التكاثر، مثل التعميم بين الأنواع المختلفة، وتفاوت جودة الفيديو، والتباين الطبيعي في العينات البيولوجية. كما تتناول الجوانب الأخلاقية لاستخدام الذكاء الاصطناعي في السياقات البيولوجية والبيطرية.

وتُظهر النتائج أن دمج تقنيات التعلم العميق في تحليل السائل المنوي يمكن أن يحسن بشكل كبير من دقة وثبات وقابلية توسيع الطرق التقليدية. ويُعد هذا العمل خطوة أولى نحو تطوير أنظمة ذكاء اصطناعي قابلة للتفسير، متعددة الأنواع، وتحافظ على خصوصية البيانات في مجال التشخيص التناسلي.

الكلمات المفتاحية: الذكاء الاصطناعي، كشف الحيوانات المنوية، YOLOv11، EfficientNet، تحليل السائل المنوي، التعلم العميق

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Dedication

To my mother,for her endless love, sacrifices, and unwavering support through every step of this journey.

To my family, for always believing in me and standing by my side with strength and encouragement.

To my friends,for their encouragement, laughter, and presence during both the challenging and joyful moments.

To my teachers and mentors,whose guidance and inspiration have helped shape my path.

To all those who have inspired and supported me along the way this work is dedicated to you.

for believing, persisting, and never giving up this achievement is a reflection of that resilience.

NASRI Houssem

To the soul of my beloved father,

whose unwavering encouragement and dreams for my future have been a guiding light in my journey. He always wished to see me one day as part of the scientific community. Though he is no longer with me, his faith continues to inspire every step I take. This work is dedicated to him, with love, gratitude, and eternal remembrance.

To my mother,

who accompanied me throughout every stage of this journey,

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And to everyone who supported me,

with a kind word, a helping hand, or a heartfelt prayer

This work is for you, with sincere appreciation and gratitude.

HANFER Zaki

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III. List of Acronyms

AI Artificial Intelligence

ML Machine Learning

DL Deep Learning

CNN Convolutional Neural Network

YOLO You Only Look Once

FPS Frames Per Second

IoU Intersection over Union

NMS Non-Maximum Suppression

GPU Graphics Processing Unit

TPU Tensor Processing Unit

UI User Interface

UX User Experience

SaaS Software as a Service

API Application Programming Interface

CSV Comma-Separated Values

mAP Mean Average Precision

CASA Computer-Assisted Sperm Analysis

ART Assisted reproductive technologies

DNA deoxyribonucleic acid

ReLU rectified linear unit

SSD Single Shot Detector

VCL curvilinear velocity

VSL straight-line velocity

VAP average path velocity

DWT discrete wavelet transforms

CSR-DCF Channel and Spatial Reliability Discriminative Correlation Filter

YAML yet another markup language

General Introduction:

Practical and Technical Context

Semen analysis is a fundamental technique in reproductive biology, essential for diagnosing male fertility issues, supporting assisted reproductive technologies (ART), and improving animal breeding programs. Traditionally conducted manually under a microscope, this analysis involves assessing critical parameters such as sperm concentration, motility, and morphology. Despite its clinical importance, manual semen evaluation suffers from significant limitations, including subjectivity, time consumption, inter-observer variability, and limited scalability across large datasets or varied species.

To address these issues, Computer-Assisted Sperm Analysis (CASA) systems were developed to automate some aspects of evaluation. However, CASA remains limited in morphological analysis and often inaccessible due to its cost and need for technical expertise. Recent advances in Artificial Intelligence (AI), especially in computer vision and deep learning, offer a new opportunity to revolutionize semen analysis. Convolutional Neural Networks (CNNs), object detection models like YOLO, and real-time tracking algorithms like Deep SORT are capable of learning complex visual patterns, enabling highly accurate, fast, and scalable analysis of spermatozoa.

A particularly challenging and unexplored area lies in multi-species sperm classification. In veterinary medicine and comparative reproductive biology, it is crucial to distinguish sperm cells from various species such as goats, donkeys, and bees based on distinct visual features like head shape, tail curvature, and motility patterns. Traditional methods struggle with this variability, making it a compelling use case for deep learning-based automation.

Problem Statement and Motivations

Traditional methods, whether performed manually or with the help of Computer-Assisted Sperm Analysis (CASA) systems, often struggle to meet the growing demands of modern reproductive science. One of the major limitations lies in their inability to adapt to the wide range of sperm shapes and movement patterns seen across different species. Moreover, these methods typically lack the speed and scalability needed for real-time, high-throughput analysis. Human subjectivity further complicates the process, introducing potential inconsistencies in evaluation. These limitations have sparked growing interest in AI-driven approaches systems designed not just to automate semen analysis, but also to accurately detect, track, and classify sperm cells across a diverse set of animal species.

Objectives

This thesis sets out to develop a comprehensive AI-powered pipeline tailored for semen analysis. At its core, the system aims to perform three essential functions: detecting sperm cells within microscopic images or videos, tracking their motion to evaluate motility in real time, and classifying them by species using a deep learning approach capable of handling multiple categories. To support this goal, a carefully annotated dataset comprising samples from various animal species will be constructed. The performance of deep learning models will then be rigorously assessed, focusing on their accuracy, processing speed, and ability to generalize across both laboratory conditions and practical, real-world scenarios.

Chapter 1: Biological Background of Semen Analysis and Sperm Morphology

This chapter provides a foundational understanding of semen composition and the biological parameters critical to assessing male fertility. It explores the functional characteristics of sperm such as motility, count, and viability alongside morphological differences observed across species. The discussion also highlights the importance of sperm classification in veterinary and biological contexts and contrasts traditional evaluation methods with emerging AI-based approaches. This background sets the stage for understanding the scientific and technological motivations behind automated, species-specific sperm analysis.

1.1 Overview of semen composition

Semen is a biologically complex fluid essential for male fertility, composed of two main components: spermatozoa and seminal plasma. While sperm cells are the reproductive agents responsible for fertilization, they represent only a small fraction of the total semen volume. The majority of the fluid is seminal plasma, a nutrient-rich medium that plays a key role in supporting sperm viability, motility, and function [1].

This plasma is not secreted by a single organ but is the result of combined contributions from multiple glands within the male reproductive system. The seminal vesicles are responsible for the largest portion of the fluid, producing a secretion that includes fructose (the primary energy source for sperm), prostaglandins, and amino acids. The prostate gland adds further components such as zinc, citric acid, and enzymes that help regulate pH and protect sperm from oxidative stress. Testicular and epididymal fluids contribute mature sperm cells, while the bulbourethral glands secrete mucus that assists in lubrication and neutralizing any acidic residues in the urethra [1].

From a biochemical perspective, seminal fluid contains both organic and inorganic substances. The organic part includes enzymes, sugars, and proteins that serve various roles, from activating sperm motility to aiding fertilization. The inorganic fraction consists of trace elements such as zinc, calcium, magnesium, and copper, each of which has been linked to specific aspects of sperm health. For instance, zinc is essential for maintaining sperm membrane integrity and DNA stability, while calcium is critical for initiating the sperm's motility [1].

Physicochemical properties such as pH, volume, viscosity, and liquefaction time are also important. Typically, semen has a slightly alkaline pH (around 7.2–7.8), which helps counter the acidic environment of the vaginal tract. Normal volume ranges from 2 to 5 milliliters per ejaculate, and the semen's ability to liquefy within a specific time frame after ejaculation is vital for the free movement of sperm [1].

In summary, semen is far more than just a carrier for sperm , it is a carefully balanced medium that plays multiple roles in ensuring successful reproduction. Understanding its composition is fundamental for evaluating male fertility and guiding both clinical and research-based assessments of reproductive health [1].

1.2 Functional sperm parameters

Semen quality is fundamentally assessed through three core parameters: sperm motility, sperm count, and viability all of which are critical indicators of male fertility. Among these, motility is often considered one of the most predictive of a sperm cell's fertilization capability.

Motility refers to the ability of sperm to move efficiently through the female reproductive tract, and it is typically evaluated in terms of the percentage of sperm that are motile, their velocity, and the quality of their movement. In clinical settings, this is often broken down into total motility, progressive motility (movement in a straight or large circular path), and non-progressive motility (movement without meaningful progression). According to the study, reduced motility was significantly observed in older men, and this reduction was closely linked to elevated levels of specific elements in the sperm, notably calcium and copper [2].

The sperm count, another essential parameter, denotes the concentration of sperm in a given volume of semen. Although both young and old men in the study provided samples within standard volume ranges, differences in elemental composition seemed to correlate more directly with motility and DNA integrity than with count alone [2].

Viability, while not the direct focus of this particular study, is closely tied to both motility and the biochemical environment in which the sperm function. High levels of DNA fragmentation, which were more frequent in older men, indicate compromised sperm viability and have been associated with impaired fertilization potential. The presence of high concentrations of elements like sulfur in seminal plasma was also shown to be negatively correlated with motility and positively associated with chromosomal damage, suggesting a broader impact on sperm health [2].

The study employed computer-assisted semen analysis (CASA) to ensure precise measurement of motility characteristics. Sperm samples were analyzed for average speed, directionality, and progression patterns, revealing that elemental imbalances particularly high intracellular calcium were linked to impaired movement patterns. This supports existing evidence that calcium plays a regulatory role in sperm flagellar motion and hyperactivation, which are necessary for successful fertilization [2].

In summary, the data reaffirm the importance of analyzing functional sperm parameters as part of a comprehensive assessment of male fertility. They also suggest that elemental composition especially in the context of aging can significantly influence motility and genomic stability. This insight opens up new possibilities for diagnostic tools and therapeutic strategies focused on improving sperm function through targeted nutritional or medical interventions [2].

1.3 Interspecies Sperm Morphology

Sperm morphology the structural design of sperm cells varies widely across species, reflecting adaptations to specific reproductive strategies and environments. Examining these differences offers insights into the evolutionary pressures shaping reproductive success.

- **Honey Bee**

Honey bee sperm are notably elongated and filamentous, measuring approximately 250–270 µm in length and about 0.7 µm in width. The head region is narrow, around 10 µm long and 0.4 – 0.5 µm wide, comprising a conical acrosomal vesicle and a linear nucleus. The flagellum features a 9+9+2 microtubular arrangement, accompanied by asymmetrical mitochondrial derivatives and accessory bodies, facilitating motility essential for successful fertilization [3].

- **Donkey**

Donkey sperm exhibit a typical mammalian structure with an oval head, midpiece, and tail. Morphological assessments have identified various abnormalities, including rough mid-pieces, macrocephaly, bent mid-pieces, and coiled tails. Despite these anomalies, a significant proportion of sperm maintain normal morphology, with studies reporting over 72% normal forms in healthy individuals [4].

- **Goat**

In goats, sperm morphology is a critical factor in fertility. Normal sperm display an oval head, intact acrosome, and a straight tail. Abnormalities such as double heads, detached acrosomes, bent tails, and cytoplasmic droplets can impair fertility. Environmental stressors, particularly high temperatures, have been linked to increased morphological defects, emphasizing the importance of managing breeding conditions [5].

- **Human**

Human sperm are characterized by an oval head measuring 5–6 µm in length and 2.5–3.5 µm in width, a midpiece packed with mitochondria, and a long tail for propulsion. Morphological assessment is a standard component of semen analysis, with normal morphology defined by the presence of at least 4% of sperm with standard structure. Deviations in morphology can affect fertility, although the exact impact varies among individuals [8].

- **Rabbit**

Rooster sperm are filiform, featuring a slender, elongated head and a long tail adapted for rapid movement through the female reproductive tract. The head is narrow and tapers into the tail, facilitating swift transit to the site of fertilization. Morphological assessments using various staining techniques have been employed to evaluate sperm quality and identify abnormalities [7].

- **Ram**

Ram sperm exhibit variability in head shape, including differences in area, perimeter, length, and width. Research has identified distinct subpopulations based on head morphology, with certain shapes correlating with higher fertility rates. This diversity within ejaculates indicates that sperm morphology plays a role in reproductive success and may be subject to selective pressures [9].

- **Rooster**

Rooster sperm are filiform, featuring a slender, elongated head and a long tail adapted for rapid movement through the female reproductive tract. The head is narrow and tapers into the tail, facilitating swift transit to the site of fertilization. Morphological assessments using various staining techniques have been employed to evaluate sperm quality and identify abnormalities [10].

- **Sea Urchin**

Sea urchin sperm are streamlined, featuring a conical head with a small acrosome, a compact midpiece, and a thin tail. Studies have revealed significant variation in sperm morphology among individuals and populations, suggesting that environmental factors and selection pressures influence these traits. Such diversity may affect fertilization success in the external spawning environment typical of sea urchins [6].

1.4 Veterinary and biological motivations for sperm species classification

Accurate assessment of sperm functionality plays a critical role in both wildlife conservation and animal breeding programs. In wildlife species, reproductive success is often hindered by limitations in evaluating semen quality using standardized, domesticated-species protocols. Many traditional methods, such as those used in domestic livestock, may not capture the physiological nuances of sperm from wild species. Thus, there is a growing demand for species-specific analytical approaches that account for unique sperm morphologies, metabolic characteristics, and environmental adaptations [11].

Moreover, in the field of conservation biology, functional sperm evaluation is essential not only for determining fertility potential but also for guiding assisted reproductive techniques (ART) such as artificial insemination (AI) and cryopreservation. Since many wildlife species are threatened or endangered, effective sperm analysis methods can significantly aid in sustaining genetic diversity through breeding programs. However, the absence of tailored analytical frameworks often results in misinterpretation of sperm quality, leading to poor outcomes in reproduction strategies [11].

In the context of domestic species, particularly equids, both interspecific (between species) and intraspecific (within species) artificial insemination efforts have revealed the critical importance of identifying and classifying sperm accurately. Differences in sperm morphology, viability, and motility patterns among species such as horses, donkeys, and mules present significant challenges during insemination. For example, sperm from donkey stallions show different kinetic and structural properties compared to horse sperm, which can influence fertilization rates when using cross-species AI techniques [12].

Accurate classification of sperm by species, therefore, becomes a crucial step in ensuring the compatibility of reproductive material and improving the efficiency of artificial insemination protocols. Furthermore, these distinctions are not just biologically relevant but also have practical implications in preserving endangered breeds, enhancing livestock productivity, and preventing interspecies reproductive failure.

In summary, both in wildlife conservation and domestic animal reproduction, there is a strong biological and veterinary imperative to develop sperm classification systems that are tailored to species-specific traits. This highlights the broader need for advanced, automated tools potentially AI-driven that can differentiate sperm by species with greater precision and reliability.

1.5 Traditional vs. AI-based species classification

Historically, the classification of sperm cells across different species has been predominantly manual, relying on microscopic examination to assess morphological features such as head shape, midpiece size, and tail length. This method is labor-intensive and subject to observer variability, which can lead to inconsistencies in results. Moreover, manual analysis is time-consuming and may not be feasible for processing large datasets, limiting its applicability in high-throughput settings [13].

The advent of artificial intelligence (AI) and deep learning technologies has introduced automated methods for sperm classification, enhancing accuracy and efficiency. AI algorithms, particularly convolutional neural networks (CNNs), can be trained on large datasets to recognize complex patterns in sperm morphology that may be challenging for human observers to discern. These systems reduce the reliance on manual examination, offering consistent and objective analysis across various species [14].

AI-based classification systems offer several advantages over traditional methods, including increased speed, scalability, and the ability to handle large volumes of data with minimal human intervention. These systems can adapt to variations in image quality and biological differences among species, providing robust and reliable classification outcomes. Furthermore, the use of AI minimizes human error and enhances reproducibility in sperm analysis [15].

While AI technologies have significantly improved sperm classification, challenges remain, such as the need for large, annotated datasets to train algorithms effectively. Additionally, the integration of AI systems into existing laboratory workflows requires careful consideration of technical and ethical aspects, including data privacy and the interpretability of AI decisions. Ongoing research aims to address these challenges and further refine AI applications in reproductive biology [16].

1.6 Conclusion

The detailed examination of semen composition and sperm functionality across species underscores the complexity inherent in reproductive biology. Understanding these biological nuances is essential for accurate semen analysis and effective fertility assessment. This chapter highlights that while traditional methods have laid important groundwork, their limitations in addressing species diversity and evaluation consistency point toward the necessity for innovative AI-driven solutions. These insights form the basis for developing automated systems that can improve accuracy and efficiency in sperm classification, particularly in veterinary and biological applications.

Chapter 2: State of the Art in AI for Biomedical and Veterinary Image Analysis

Artificial Intelligence has revolutionized the landscape of biomedical and veterinary imaging, offering powerful tools to extract meaningful insights from complex visual data. This chapter explores how AI, particularly deep learning and convolutional neural networks (CNNs), is reshaping image analysis in the fields of biology and medicine. Beginning with a foundational overview of AI applications in life sciences, the chapter delves into the architecture and capabilities of CNNs and their transformative impact on medical imaging tasks, such as disease detection and cellular classification [20].

Special attention is given to AI's role in microscopy, where it enhances image clarity, automates analysis, and enables the precise identification of biological structures. The chapter further investigates object detection and classification techniques as applied to biological datasets, laying the groundwork for understanding modern sperm analysis systems such as CASA and OpenCASA. Finally, it addresses the growing need for multi-species image classification and tracking, highlighting recent advancements and persisting challenges. By identifying gaps in current methodologies, this chapter sets the stage for future innovations in AI-driven reproductive diagnostics [20].

2.1 Introduction to Artificial Intelligence in biology and medicine

Artificial Intelligence is increasingly reshaping the landscape of biological research by enabling the rapid analysis of large-scale, complex biological data. In areas such as genomics and proteomics, AI techniques, particularly machine learning, help uncover intricate biological patterns that are often difficult to detect using classical analytical approaches. These methods support advancements in predicting protein structures, interpreting gene expression, and mapping cellular pathways, thereby improving our understanding of various biological systems and disease processes [20].

In the medical domain, AI is revolutionizing diagnostics and treatment planning. Machine learning models can analyze medical images, electronic health records, and clinical data to assist in early disease detection, prognosis, and personalized treatment strategies. For instance, AI algorithms have been developed to detect abnormalities in imaging studies, predict patient outcomes, and recommend tailored therapeutic interventions, thereby enhancing clinical decision-making and patient care [17].

AI is also making significant strides in drug discovery and development. By analyzing chemical structures, biological data, and clinical trial results, AI models can identify potential drug candidates, predict their efficacy, and optimize clinical trial designs. This accelerates the drug development process and reduces associated costs. Notably, AI-driven platforms have been instrumental in identifying novel compounds and repurposing existing drugs for new therapeutic uses [18].

Despite the promising applications of AI in biology and medicine, several challenges persist. These include data privacy concerns, the need for large and diverse datasets to train models effectively, and the interpretability of AI decisions. Ethical considerations, such as ensuring unbiased algorithms and addressing potential disparities in healthcare access, are also critical. Ongoing research aims to develop explainable AI models and establish regulatory frameworks to address these issues [19].

2.2 Deep Learning and Convolutional Neural Networks (CNNs)

Deep learning is a specialized subset of machine learning focused on algorithms modeled after the structure and function of the human brain, particularly artificial neural networks. These models consist of multiple layers that progressively extract higher-level features from raw input data. Unlike traditional machine learning approaches that often rely on manual feature engineering, deep learning models can automatically learn representations that are highly effective for classification, segmentation, and detection tasks. This makes them particularly suitable for complex data modalities such as images and biomedical signals [21].

2.2.1 Architecture of Convolutional Neural Networks

Convolutional Neural Networks (CNNs) are the most widely used deep learning models in image recognition and analysis. A CNN typically comprises several layers: convolutional layers that apply learnable filters to the input image, activation layers (e.g., ReLU) that introduce non-linearity, pooling layers that reduce spatial dimensions, and fully connected layers that perform classification or regression tasks. This layered design allows CNNs to learn spatial hierarchies of features, starting from edges and textures in lower layers to more complex shapes in higher ones [22].

CNNs are inherently suited for visual pattern recognition because they preserve spatial relationships in data and reduce the number of trainable parameters compared to fully connected networks. They can identify and differentiate between subtle visual cues, such as morphological variations in cell structures or anatomical features. Their robustness to translation, scaling, and even some degree of rotation makes them particularly useful in biological image datasets where sample variability is high [23].

Training deep CNNs requires large datasets and computational resources. To prevent overfitting and improve generalization, regularization techniques like dropout (randomly disabling neurons), weight decay (penalizing large weights), and data augmentation (altering images to simulate variability) are widely used. Moreover, optimization algorithms such as Adam and RMSProp are employed to update weights efficiently during training, helping models converge faster and more stably [21].

2.2.2 CNNs in Biological and Medical Imaging

In medical and biological contexts, CNNs have been applied successfully to detect tumors, classify retinal diseases, identify cells in histopathological images, and segment organs in 3D scans. Their ability to process high-resolution data and learn discriminative features without handcrafted inputs has made them a go-to solution in biomedical image analysis. CNNs can match or even surpass human performance in certain diagnostic tasks, providing reliable assistance in clinical workflows [25].

2.3 AI in microscopy and biomedical image processing

The introduction of AI techniques, especially deep learning, has significantly transformed the way microscopy data is analyzed. These systems are capable of learning complex visual patterns directly from labeled image datasets, enabling them to classify and detect cells, organelles, and anomalies in biomedical samples. Compared to manual methods, AI offers higher consistency and the ability to process large image volumes rapidly, which is particularly beneficial for high-throughput biological research. This shift allows researchers to focus more on interpretation rather than manual data handling, enhancing both efficiency and reproducibility in scientific workflows [26].

Biomedical microscopy images often suffer from various types of noise and inconsistencies that can impair analysis. Deep learning models, particularly those trained to perform image restoration tasks, have been shown to significantly improve image quality. These methods can enhance contrast, reduce background noise, and recover image details lost during acquisition. In particular, models such as convolutional autoencoders have been successfully used to learn how to map low-quality images to higher-quality counterparts, making them valuable in applications where precise image details are critical, such as fluorescence microscopy [27].

Accurate segmentation of microscopic structures is critical in biomedical research, for tasks such as counting cells, quantifying biomarker expression, and identifying organelles. CNN-based models, such as U-Net, have become standard for semantic and instance segmentation in medical and biological datasets. These models provide pixel-level classification, making them ideal for identifying intricate biological features within complex images. U-Net's encoder-decoder architecture with skip connections helps preserve spatial information, enabling precise segmentation even with limited training data [28].

AI-powered image analysis is widely used in pathology, hematology, and developmental biology. For example, AI has been used to detect malaria parasites in blood smears, classify cancer subtypes in histopathological images, and track embryonic development in time-lapse microscopy. These tools support early diagnosis, disease monitoring, and large-scale screening, especially in low-resource settings where expert interpretation may be limited. Importantly, these AI systems can operate at scale, processing thousands of images with minimal human intervention [29].

Recent developments also involve the integration of AI algorithms directly into microscopy hardware. Smart microscopes equipped with onboard processing units can perform real-time analysis during image acquisition. This enables adaptive imaging, where the microscope dynamically changes parameters (e.g., focus, magnification) based on live feedback from the AI. Such real-time decision-making streamlines experimental workflows and reduces the volume of irrelevant or low-quality data acquired [30].

2.4 Object Detection and Classification in Biological Datasets

Object detection and classification have become core components of computer vision tasks in biological and medical imaging. These techniques are essential for tasks such as identifying cells, locating organs, and segmenting structures in microscopy or radiological images. In biological contexts, object detection allows for the identification and localization of cells, bacteria, or subcellular structures, often under conditions of poor contrast, noise, and complex morphology. Classification, on the other hand, enables automated decisions about the identity or state of the detected objects distinguishing between healthy and pathological tissue, cell types, or even species-specific sperm cells [28].

In recent years, deep learning techniques, particularly Convolutional Neural Networks (CNNs), have significantly improved object detection and classification accuracy. CNN-based detectors like YOLO (You Only Look Once), SSD (Single Shot Detector), and Faster R-CNN have been adapted to biomedical applications. These architectures are trained on annotated datasets to detect biological objects with high precision and can be fine-tuned on domain-specific data, making them particularly suitable for diverse biological tasks [28].

For instance, U-Net, with its architecture shown in Figure 01, has become a standard for cell and tissue detection. It has been successfully applied to the segmentation of neurons, blood vessels, and other microscopic features in biomedical datasets. Its ability to perform well on small datasets makes it particularly useful in research domains where data collection is expensive or time-consuming. [28].

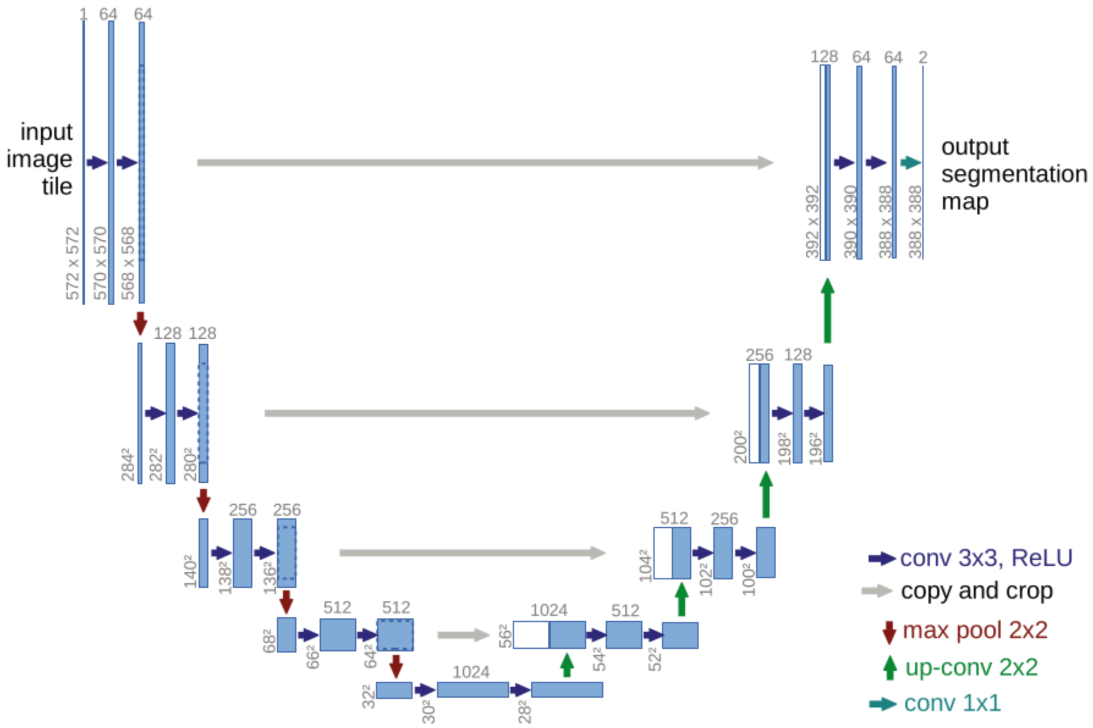


Figure 01 : U-Net: Convolutional Networks for Biomedical Image Segmentation [28]

In parallel, label-free approaches that do not require staining or tagging biological samples are gaining traction due to their ability to preserve sample integrity. In this domain, deep learning, combined with photonic time-stretch imaging, enables accurate classification of single cells in a label-free and high-throughput manner. This method extracts high-dimensional biophysical features from optical phase and intensity images and feeds them into a supervised deep neural network for classification. Such approaches offer significant potential for large-scale, non-invasive biological diagnostics [31].

2.5 Review of Sperm Analysis Systems (CASA, OpenCASA, etc.)

Computer-Assisted Sperm Analysis (CASA) systems have significantly advanced the evaluation of sperm quality by providing objective, reproducible, and quantitative assessments. Unlike manual evaluations, which are susceptible to observer bias and variability, CASA systems utilize automated techniques to measure parameters such as sperm concentration, motility, and morphology. These systems have become integral in both clinical and research settings for assessing male fertility [32].

Traditional Methods in CASA for Sperm Detection and Tracking:

Traditional CASA systems primarily rely on image processing techniques to detect and track spermatozoa. The process begins with capturing microscopic images or videos of sperm samples using phase-contrast or bright-field microscopy. These images are then analyzed using algorithms that enhance contrast, reduce noise, and segment sperm cells from the background. One common approach involves the use of thresholding methods to differentiate sperm heads from the background. After segmentation, morphological operations are applied to refine the detection, ensuring that only objects matching the expected size and shape of sperm heads are considered. This process helps in accurately identifying individual sperm cells for further analysis [33].

For tracking sperm motility, CASA systems analyze the movement and path of sperm heads across sequential frames. By calculating the displacement of each sperm head over time, the system derives various kinematic parameters such as curvilinear velocity (VCL), straight-line velocity (VSL), and average path velocity (VAP). As illustrated in Figure 02, these parameters are obtained by tracking sperm centroids across frames to generate different trajectory types. They are crucial for assessing the motility characteristics of spermatozoa. [33].

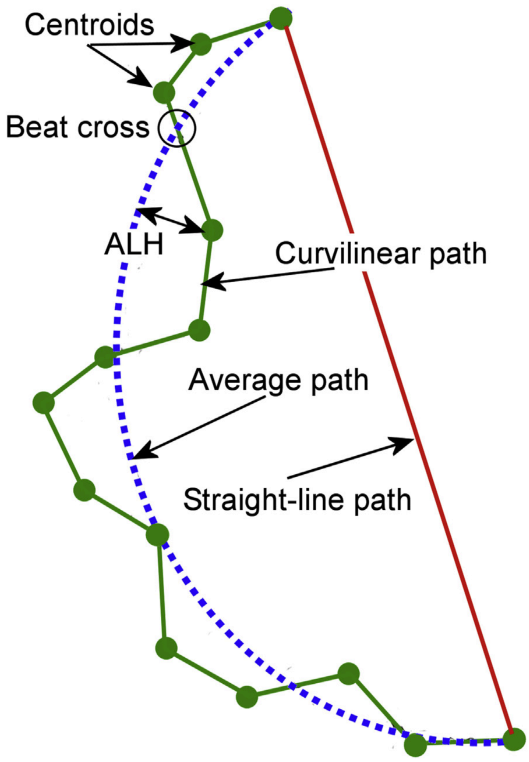


Figure 02 :Illustration of CASA terminology, depicting how sperm centroids are tracked across frames to generate curvilinear, average, and straight-line trajectories, from which metrics such as VCL, VAP, VSL, ALH, and BCF are derived[33]

These parameters are mathematically defined to ensure consistency and reproducibility across studies.

Curvilinear Velocity (VCL): is calculated as the time-averaged speed along the actual path of the sperm cell [63]:

$$VCL = \frac{1}{p}\sum_{i=1}^{k-1}\frac{\sqrt{(x_{i+1}-x_i)^2+(y_{i+1}-y_i)^2}}{\Delta t}$$

Straight-Line Velocity (VSL): represents the net distance between the starting and end point divided by time [63]:

$$VSL = \frac{\sqrt{(x_k-x_1)^2+(y_k-y_1)^2}}{p}$$

Average Path Velocity (VAP): is the velocity along a smoothed average path [63]:

$$VAP = \frac{1}{p}\sum_{i=1}^{k-1}\frac{\sqrt{(x_{avg,i+1}-x_{avg,i})^2+(y_{avg,i+1}-y_{avg,i})^2}}{\Delta t}$$

- (x_i, y_i) are the coordinates of the sperm head in each frame
- k = total number of points in one sperm trajectory
- p = number of frames
- Δt = time interval between frames

However, traditional image processing methods face challenges, especially in samples with high sperm density or poor image quality. Overlapping sperm cells, debris, and variations in lighting can lead to inaccuracies in detection and tracking. To address these issues, some CASA systems incorporate additional preprocessing steps, such as filtering and morphological analysis, to enhance detection accuracy [33].

Filtering Techniques to Eliminate Debris and Artifacts :

To improve the accuracy of sperm detection and tracking, various filtering techniques have been employed in CASA systems to eliminate debris and artifacts. Median filtering is commonly used to reduce noise while preserving edges, which is essential for maintaining the integrity of sperm morphology during analysis [16]. Gaussian filters are also applied to smooth images and suppress random noise, enhancing the visibility of sperm cells against the background [34].

Additionally, discrete wavelet transforms (DWT) have been utilized for noise reduction and feature extraction, allowing for better differentiation between sperm cells and background noise [34].In some cases, glare filters and diffraction correction techniques are implemented to address optical artifacts arising from microscopy, further improving image quality for accurate analysis [16].

Advancements and Integration with Open-Source Tools :

Recognizing the limitations of traditional methods, recent developments have focused on integrating advanced algorithms and open-source tools into CASA systems. For instance, OpenCASA and similar platforms offer customizable frameworks that allow researchers to implement and test new detection and tracking algorithms. As shown in Figure 03, these tools often include modules for preprocessing, segmentation, and kinematic analysis, providing a flexible environment for sperm analysis. [35].

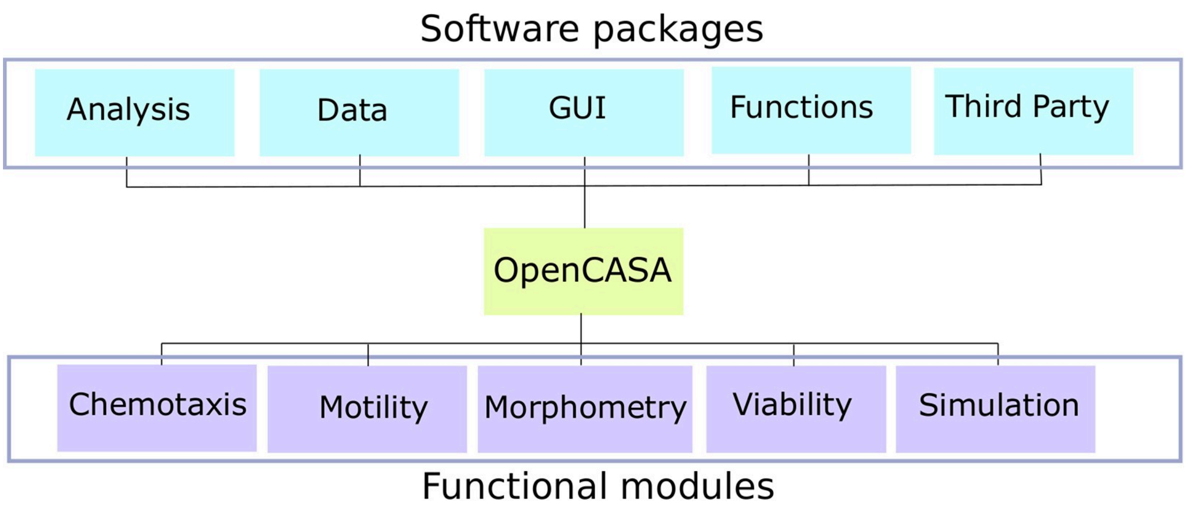


Figure 03 : OpenCASA overview diagram [35]

2.6 Literature review on multi-species image classification and tracking

As AI applications expand in reproductive and veterinary sciences, the ability to classify and track sperm from different animal species has become increasingly important. This section reviews key studies focused on multi-species image analysis, highlighting the methods used to detect, differentiate, and monitor sperm cells across various species. It also explores the challenges involved, such as differences in morphology and movement patterns, and the solutions proposed through machine learning and deep learning techniques [36].

2.6.1 Advancements in CASA Systems for Multi-Species Analysis

Computer-Assisted Sperm Analysis (CASA) systems have evolved to accommodate the analysis of semen samples from various species, including humans, domestic animals, and wildlife. This adaptability is crucial for both clinical diagnostics and research in reproductive biology. Modern CASA systems are designed to provide consistent and objective measurements across different species, ensuring reliable assessments of sperm motility and morphology [36].

2.6.2 Challenges in Multi-Species Sperm Analysis

Analyzing sperm from different species presents unique challenges due to variations in sperm size, shape, and motility patterns. Traditional image processing techniques may struggle with these differences, leading to inaccuracies in detection and tracking. To address this, researchers have developed simulation tools that generate synthetic semen images with controllable parameters, allowing for the testing and validation of CASA algorithms under various conditions [37].

2.7 Summary and Gap Identification

Over the past decades, substantial advancements have been made in the field of sperm analysis and biomedical image processing. Classical Computer-Assisted Sperm Analysis (CASA) systems have laid the groundwork for automated sperm tracking by leveraging traditional image processing techniques, such as thresholding, morphological operations, and object tracking algorithms. These tools have improved the objectivity and reproducibility of DCNN sperm quality assessments across species, particularly in clinical and veterinary contexts [33].

Recent advancements in deep learning, particularly the application of convolutional neural networks (CNNs), have significantly improved the analysis of sperm motility in semen samples. For instance, a study by Sætra et al. demonstrated the effectiveness of a deep convolutional neural network model, specifically ResNet-50, in classifying sperm motility into WHO-defined categories. The model achieved a mean absolute error (MAE) of 0.05 in predicting progressive, non-progressive, and immotile spermatozoa, indicating a high level of accuracy in automated semen analysis [39].

Additionally, Mohammadi et al. proposed a method combining deep learning with a modified CSR-DCF tracking algorithm to detect and track sperm cells in phase-contrast microscopy images. Their approach achieved an average precision of 99.1% in detection and an F1 score of 96.61% in tracking, highlighting the potential of integrating deep learning techniques for precise sperm motility analysis [38].

Additionally, while deep learning has demonstrated promise, the reliance on large, labeled datasets remains a bottleneck. Publicly available datasets for sperm analysis, especially across different species, are limited. Moreover, interpretability of deep learning models in biomedical applications continues to be an issue, which hinders clinical adoption [37].

Therefore, there is a clear need for unified, intelligent systems that combine the strengths of classical CASA algorithms with modern AI techniques capable of generalizing across species, reducing annotation burden through self-supervised learning, and offering explainable insights to end-users. This integrative direction holds the potential to overcome current limitations and drive the next generation of automated sperm analysis platforms.

2.8 Conclusion

Chapter 2 has provided a comprehensive overview of the current landscape of artificial intelligence in biomedical and veterinary image analysis. It began by contextualizing the role of AI,particularly deep learning,in revolutionizing biological and medical research. A focus on convolutional neural networks (CNNs) highlighted how their architecture underpins most modern image recognition systems, especially in microscopy and diagnostic imaging. The chapter also examined the application of AI in detecting and classifying biological structures, including sperm cells, through systems like CASA and OpenCASA. Special attention was given to multi-species analysis, where the complexity of morphological variation poses unique challenges. By reviewing both technological advances and ongoing limitations, this chapter sets the stage for targeted innovation in automated, species-specific sperm evaluation,pointing to existing gaps that the subsequent research aims to address.

Chapter 3: Dataset Construction and Labeling Strategy

This chapter outlines the methods and sources used for data acquisition, preprocessing, annotation, and statistical analysis for the dataset developed in this study. The data were collected to support the tasks of species classification, motility analysis, and tracking, focusing on microscopic imaging of biological specimens.

3.1 Image and Video Data Acquisition (Microscope Imaging)

The dataset was constructed using image and video samples acquired through microscope imaging from two principal sources:

- **Science Biology Lab, Université Abedrahmen Mira:** Provided static microscopy images and short video sequences for classification of biological specimens (e.g., bees, goats, donkeys) and motility analysis for human sperm. Imaging was performed using a CASA's camera

- **Clinic of Dr. Lalaoui, Bejaia:** Supplied video data for tracking and mobility analysis, focusing on human sperm motility. Videos were recorded at 30 fps using an insein li fung camera 48 Megapixel.

All samples were collected under controlled laboratory conditions. These instruments allowed for the observation of subtle motility patterns and clear morphological features essential for accurate species identification and motion analysis.

3.2 Species diversity in dataset

The dataset encompasses a diverse set of biological specimens captured under microscopic conditions, allowing for the exploration of various classification and motility patterns. This diversity is essential to ensure the generalizability of the models across different biological domains. The following categories were included:

- **Human :** Sperm samples from human donors were collected for both motility and vitality assessment. These samples serve clinical and diagnostic purposes, particularly in reproductive medicine and fertility evaluation. An example of a microscopic image of human spermatozoa is presented in Figure 04.

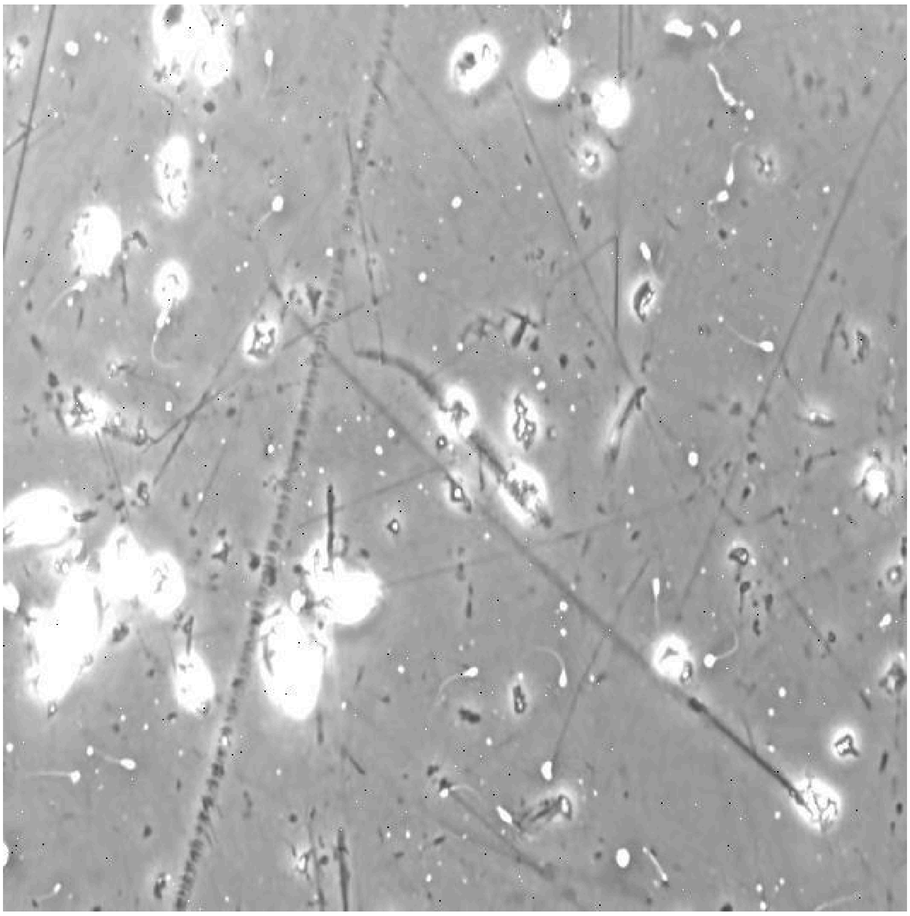


Figure 04 :Microscopic Image of Human Spermatozoa

- **Goat :** Goat spermatozoa are commonly studied in veterinary and livestock reproduction. Their motility patterns provide valuable insight into domestic animal fertility. A representative microscopic image of goat spermatozoa is shown in Figure 05.

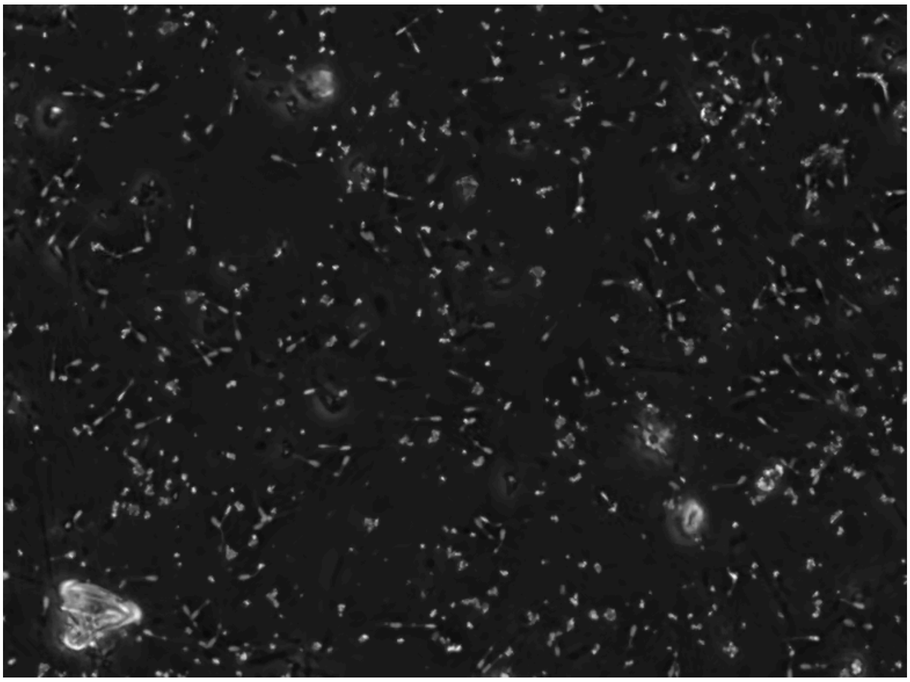


Figure 05 :Microscopic Image of Goat Spermatozoa

- **Donkey :** Donkey sperm samples were included to broaden the representation of equine species. Their larger spermatozoa and slower motility dynamics offer a contrast with smaller mammals. Figure 06 illustrates a microscopic view of donkey spermatozoa.

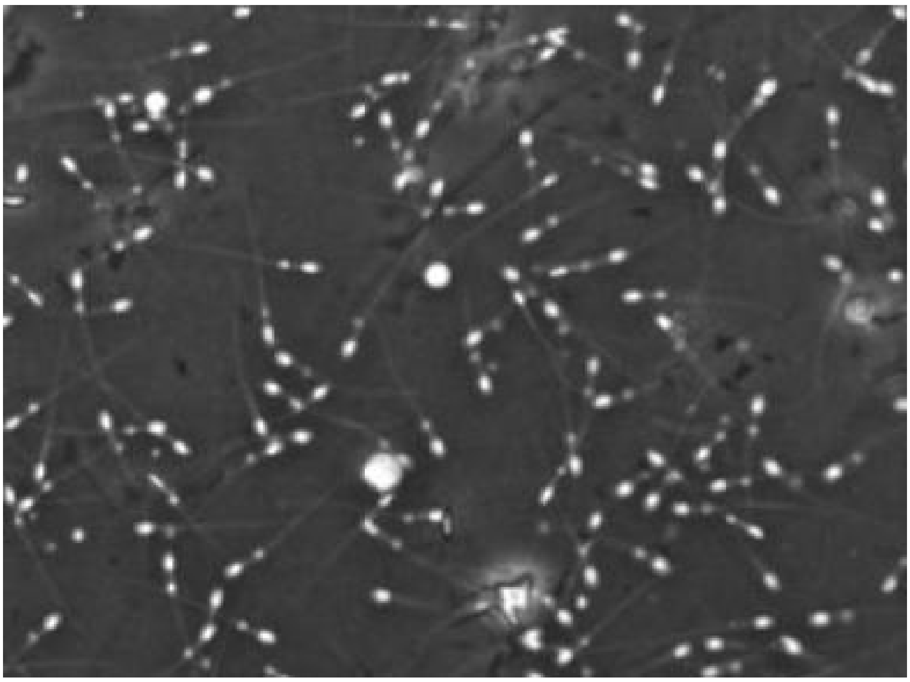


Figure 06 :Microscopic Image of Donkey Spermatozoa

- **Ram :** Ram sperm are widely used in artificial insemination research and represent an important species in ruminant reproductive studies. An example of ram spermatozoa under the microscope is displayed in Figure 07.

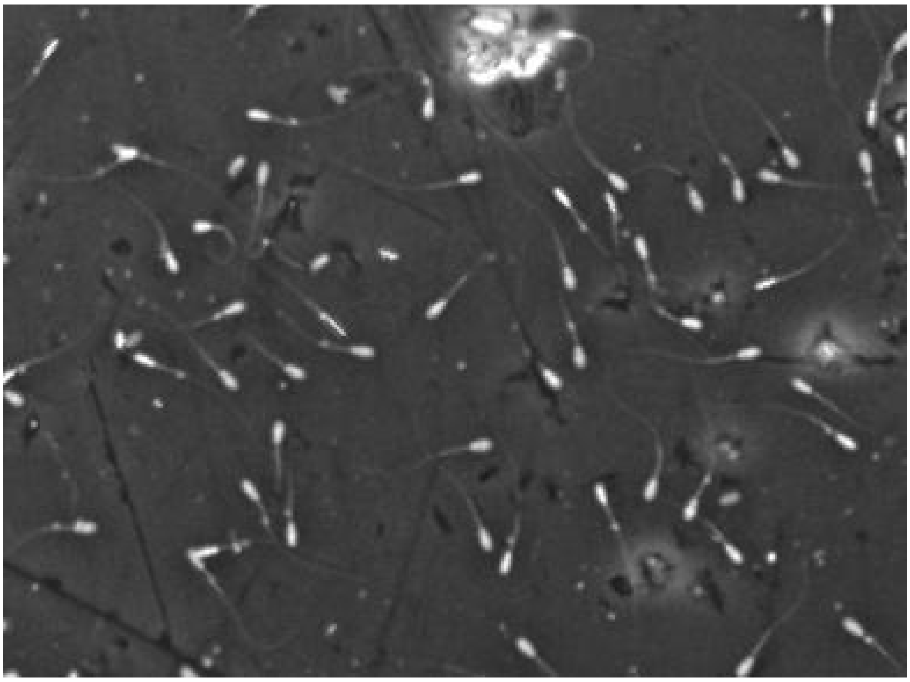


Figure 07 :Microscopic Image of Ram Spermatozoa

- **Rabbit :** Rabbit sperm samples serve as an established model in mammalian fertility research due to their ease of collection and well-characterized reproductive physiology. As seen in Figure 08, rabbit spermatozoa exhibit distinct morphological features.

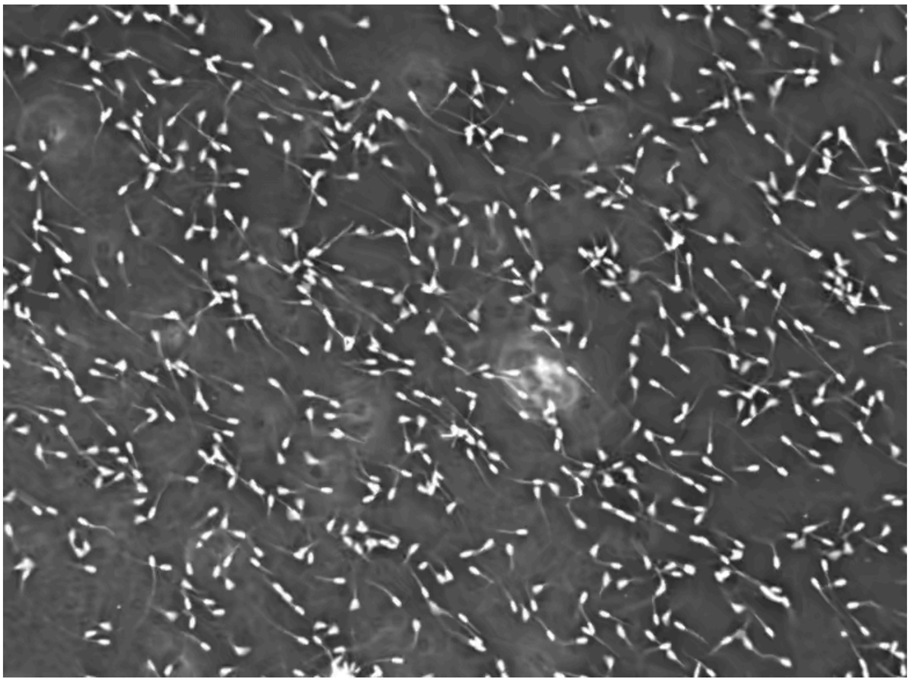


Figure 08 :Microscopic Image of Rabbit Spermatozoa

- **Rooster :** The rooster, representing avian species in the dataset, adds unique biological variation due to fundamental differences in sperm morphology and fertilization mechanisms compared to mammals. Figure 09 shows a microscopic image of rooster spermatozoa.

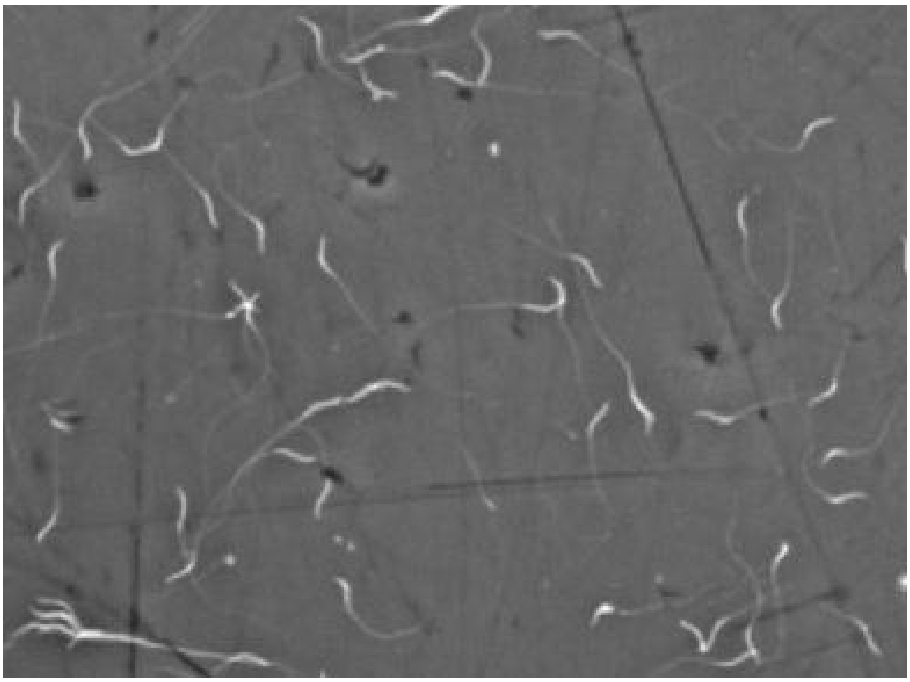


Figure 09 :Microscopic Image of Rooster Spermatozoa

- **Bee :** As an invertebrate, bee spermatozoa offer unique insights into non-mammalian motility traits. Their smaller size and distinct morphology introduce additional complexity for microscopic classification tasks. A detailed image of bee spermatozoa is provided in Figure 10.

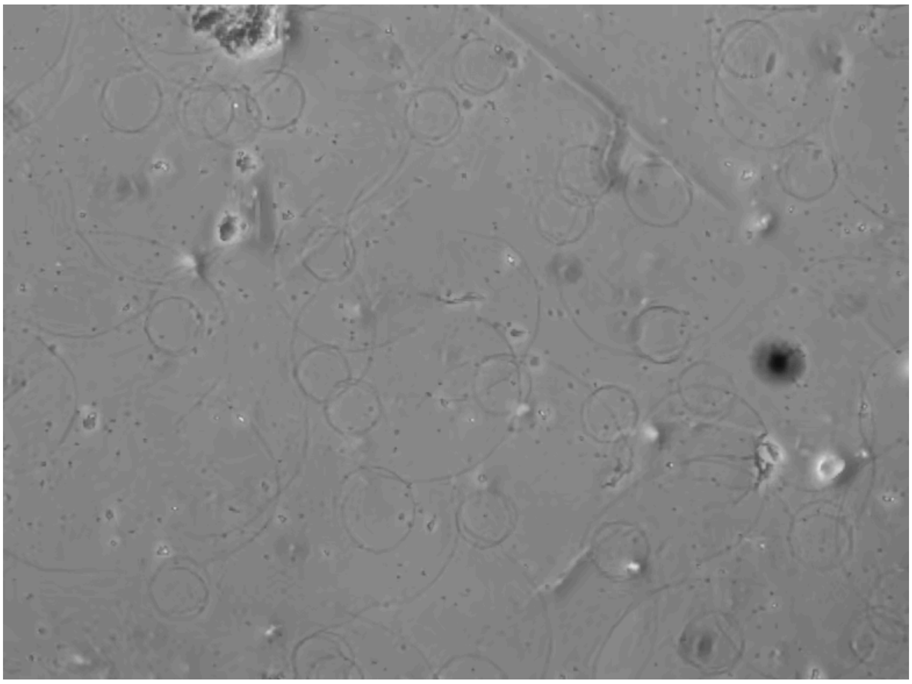


Figure 10 :Microscopic Image of Bee Spermatozoa

- **Sea Urchin :** Sea urchin sperm are frequently used in marine and developmental biology, particularly for studying fertilization under external conditions. Their flagellar motion is highly visible and regular under microscopy, making them ideal for tracking algorithms. An example of sea urchin spermatozoa can be seen in Figure 11.

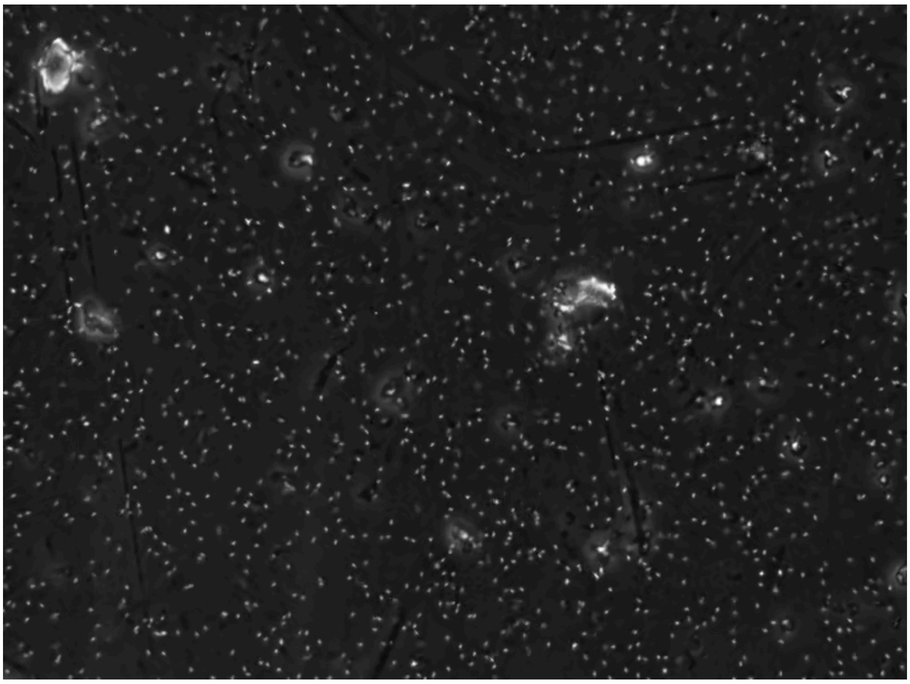


Figure 11 :Microscopic Image of Rabbit Spermatozoa

This interspecies composition enhances the dataset's applicability for training general-purpose models in sperm classification and motion tracking, while also supporting biological research into reproductive variability across taxonomic groups.

3.3 Image Preprocessing

All raw data composed of microscope videos and still images underwent a multi‑stage preprocessing pipeline to standardize formats, extract relevant frames, partition the dataset, and enhance image quality prior to annotation or model training. The workflow is summarized in Figure 12. Each step was implemented in Python using OpenCV, PIL, NumPy, and standard file‑operations modules. Below, we describe each stage.

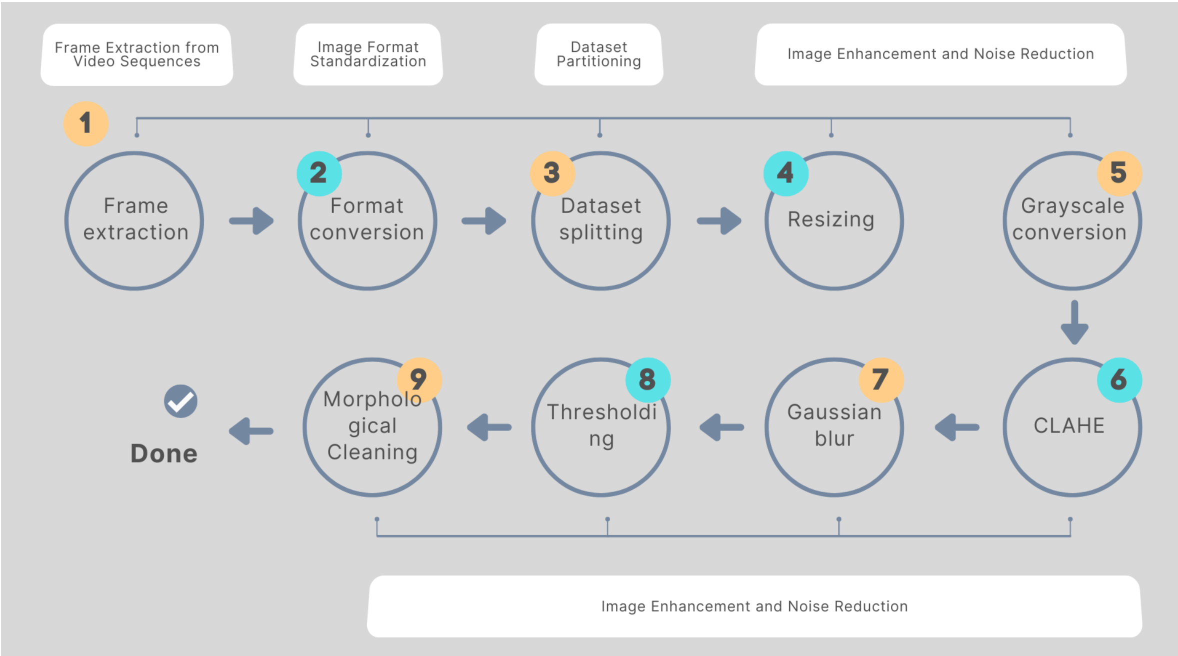


Figure 12 :Workflow Diagram of Image Preprocessing Pipeline

3.3.1 Frame Extraction from Video Sequences

Microscope videos of sperm were converted into individual still frames. This step ensured that dynamic sequences could be treated as separate images, facilitating both frame-by-frame analysis and batch processing. Each video was decomposed into its constituent frames, then saved the first and the last ones in a lossless image format to preserve fine morphological details.

3.3.2 Image Format Standardization

To maintain uniformity across the dataset, all images were converted to a single format (PNG). Many species folders initially contained JPEG files; these were transformed into PNG to avoid compression artifacts and to guarantee compatibility with annotation and training tools. By standardizing the image format, subsequent processing steps could operate on a homogeneous set of files.

3.3.3 Dataset Partitioning

Once all frames and still images were collected and standardized, the dataset was divided into separate directories for training (80 %) and validation (20 %). Images were shuffled randomly before splitting to prevent any ordering bias. This partitioning strategy ensures that models can be trained on a representative subset of data while reserving an independent sample for unbiased evaluation.

3.3.4 Image Enhancement and Noise Reduction

Once frames and still images were extracted, converted, and organized, each image underwent enhancement to optimize sperm visibility and suppress noise. The following operations were applied in sequence, leveraging OpenCV (cv2) and NumPy:

- **Resizing**

All images were uniformly resized to a fixed width while maintaining aspect ratio (via cv2.resize or imutils.resize). Standardizing spatial scale ensures that sperm cells occupy consistent pixel dimensions, reducing variability across samples and lowering computational burden during feature extraction and tracking.

- **Grayscale Conversion**

The resized color images were converted to grayscale using

```
gray_img = cv2.cvtColor(resized_img, cv2.COLOR_BGR2GRAY)
```

Since color information is not essential for motility or morphology detection, this step reduces each image to a single‑channel intensity map, streamlining subsequent processing (e.g., filtering, thresholding) and lowering memory usage.

- **Contrast Enhancement via CLAHE**

To correct uneven illumination and amplify subtle sperm features, Contrast Limited Adaptive Histogram Equalization (CLAHE) was performed:

```
clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8,8))
```

```
enhanced_img = clahe.apply(gray_img)
```

CLAHE divides the image into small contextual regions (tiles) and equalizes their histograms individually, applying a clip limit to prevent overamplification of noise. This local enhancement highlights sperm heads often only marginally darker than the background and mitigates lighting gradients introduced by microscope lamps.

- **Gaussian Filtering**

Following contrast adjustment, a Gaussian blur with a 5×5 kernel was applied:

```
blurred_img = cv2.GaussianBlur(enhanced_img, (5,5), 0)
```

Convolution with a Gaussian kernel smooths the image by attenuating high-frequency noise (e.g., speckle, microscope sensor artifacts) while preserving edge structures of sperm cells. This noise reduction is essential for reliable contour detection and prevents spurious false positives in downstream detection.

- **Thresholding and Morphological Cleaning**

To produce a binary mask isolating sperm regions, adaptive or Otsu's thresholding was employed:

```
_, binary_mask = cv2.threshold(blurred_img, 0, 255, cv2.THRESH_BINARY_INV +
cv2.THRESH_OTSU)
```

Inverting the threshold ensures that sperm heads (dark regions) become white (foreground) on a black background. Subsequent morphological opening (cv2.morphologyEx) with a small structuring element (e.g., 3×3 kernel of ones) removed isolated speckles and smooths object contours:

```
kernel = np.ones((3,3), np.uint8)
clean_mask = cv2.morphologyEx(binary_mask, cv2.MORPH_OPEN, kernel)
```

This opening operation (erosion followed by dilation) eliminates small artifacts and closes minor gaps within sperm regions, producing contiguous blobs that correspond reliably to individual sperm heads.

3.4 Annotation Strategy

Annotation was conducted using the Roboflow platform, which provided tools for efficient and scalable labeling. The labeling process was structured across three main tasks:

- **Bounding Boxes (Detection)**: Each organism was enclosed within a bounding box as shown in Figure 13, serving as input for object detection models. This spatial localization allows models to identify the position and size of each specimen.

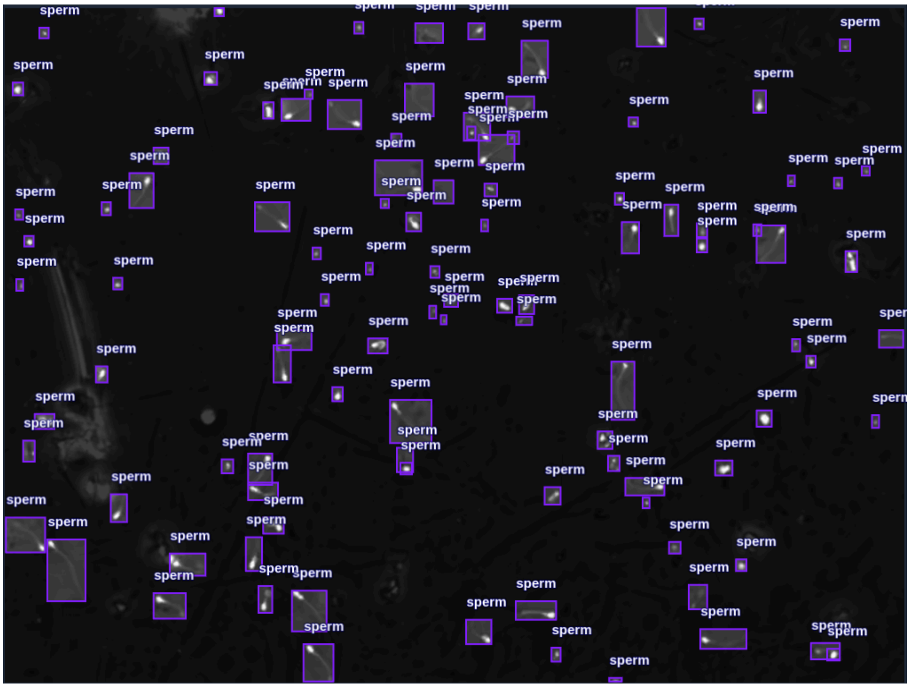


Figure 13: Example of data annotation from our dataset

- **Track IDs (Motility)**: In the context of video sequences, each organism was given a distinct Track ID to ensure consistent identification across frames. This continuity over time facilitates the analysis of motility characteristics, including speed, trajectory linearity, and oscillatory behavior. Figure 14 illustrates an example of how Track IDs were assigned during our video processing workflow.

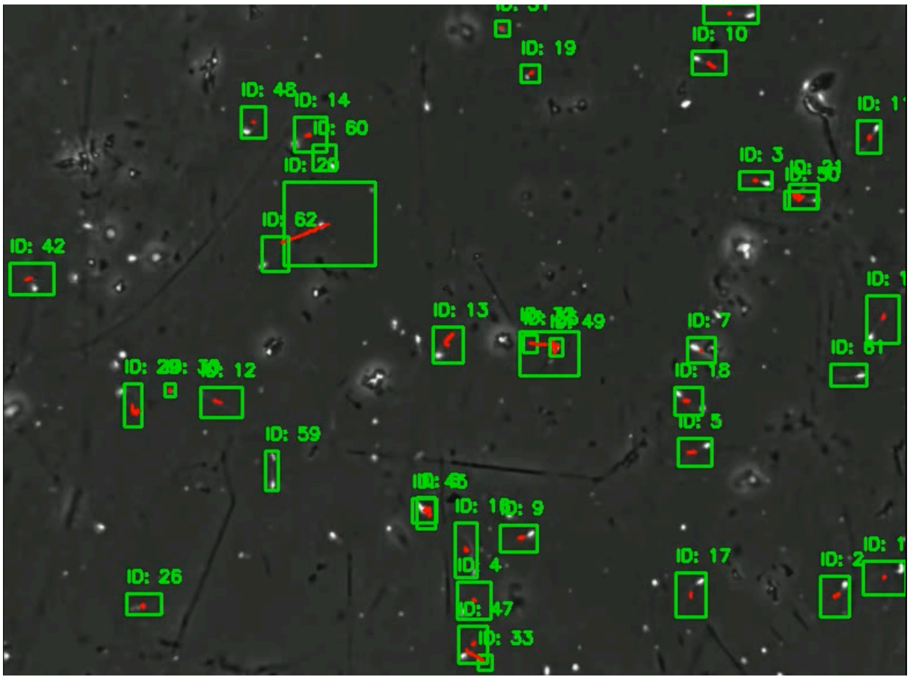


Figure 14: Example of ID assignment in our videos treatment

All annotations were reviewed to minimize labeling bias and misclassification, especially in species with subtle morphological similarities.

3.5 Class Distribution and Dataset Statistics

The final dataset comprises a total of 23670 images. The class distribution is as follows:

- **Classification Dataset**

Species	Number of Samples	Train	Validation
Human	4200	3500	700
Goat	3480	2783	697
Donkey	2000	1600	400
Ram	2000	1600	400
Rabbit	216	172	44
Rooster	2000	1600	400
Bee	3746	3046	700
Sea Urchin	6028	4822	1206
Total	23670	19123	4547

Table 01: Classification Dataset Distribution

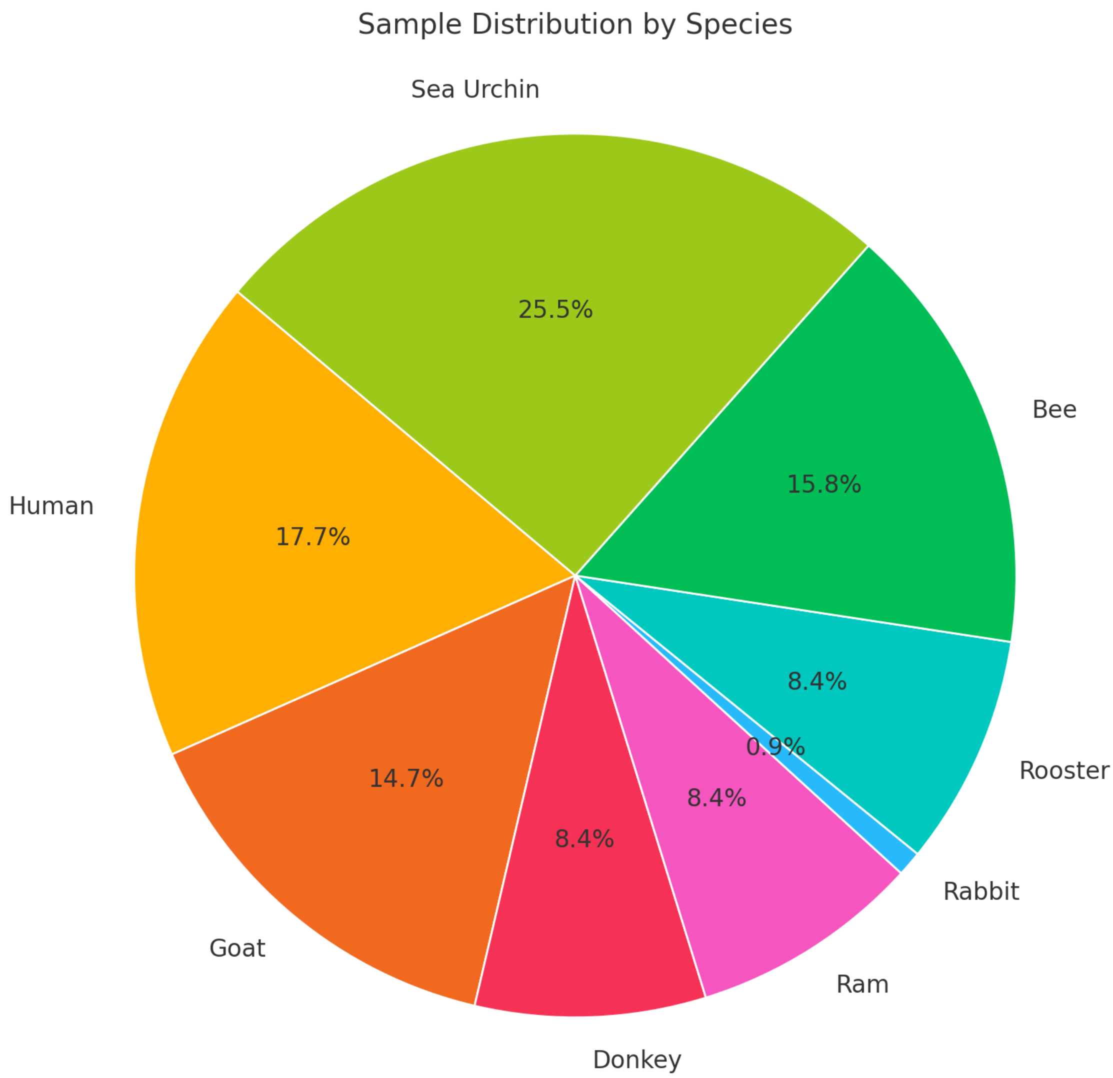


Figure 15: Pie chart illustrating the distribution of total samples across different species.

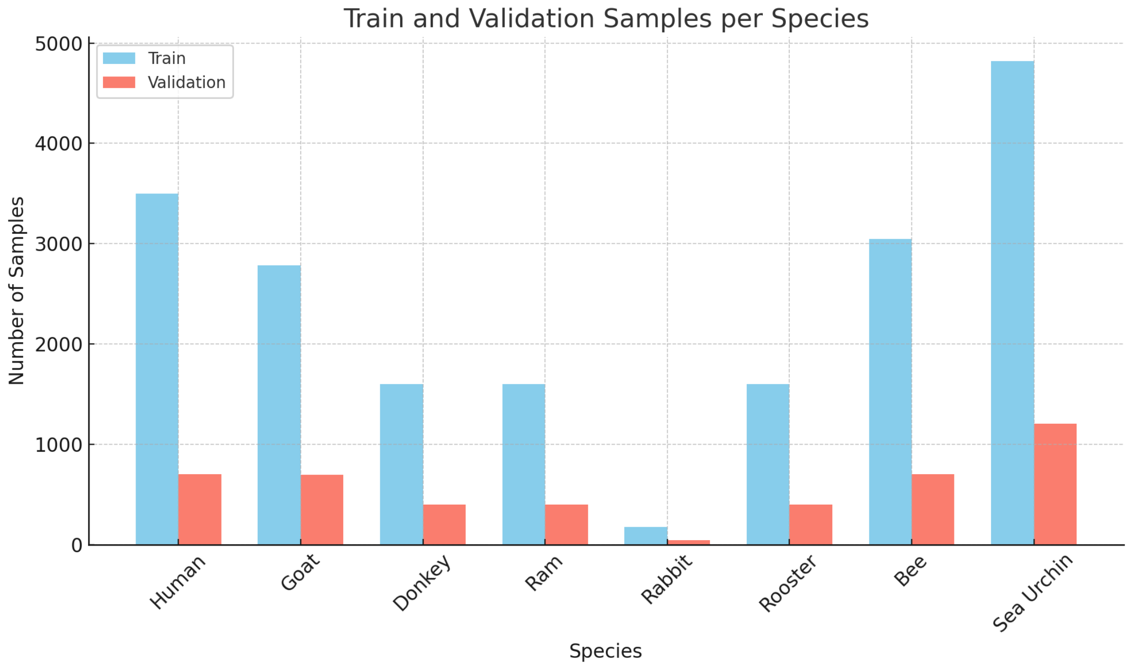


Figure 16: Grouped bar chart showing the number of training and validation samples for each species

- **Motility and Tracking Dataset**

Species	Number of Samples	Train	Validation
Human	1266	1047	219

Table 02: Detection and tracking Dataset Distribution

These statistics highlight the heterogeneity and balance of the dataset, ensuring robust training and evaluation of learning models.

3.6 Challenges in Interspecies Annotation

The annotation process presented several challenges:

- **Morphological Overlap:** Species with similar shapes or coloration at the microscopic scale, such as certain insects or small mammals, were difficult to distinguish without expert input.
- **Motion Blur**: In video sequences, fast-moving organisms occasionally appeared blurred, complicating both detection and tracking.

- **Scale Variability:** Differences in specimen size across species introduced bias in bounding box dimensions, requiring normalization strategies during model training.
- **Partial Occlusion:** Some samples showed organisms partially outside the field of view or overlapping, making it challenging to assign correct labels and track continuity.

To mitigate these issues, annotation guidelines were developed and consistently applied, and edge cases were flagged for expert review. Moreover, post-processing scripts were used to validate consistency in Track IDs and class labels across frames.

Chapter 4: Methodology for Detection, Tracking, and Classification

This chapter outlines the methodology followed to develop an AI-based pipeline for semen analysis, focusing on three core components: detection, tracking, and classification of spermatozoa. It begins with the problem formulation and justification for model choices, then details the selection and configuration of detection and classification models. Feature engineering strategies and training protocols are also discussed, along with a comparison between multi-task learning and modular pipeline approaches. The aim is to build a robust system capable of accurate and efficient analysis across diverse scenarios.

4.1 Problem formulation

Accurate analysis of sperm cells plays a critical role in reproductive biology, fertility diagnostics, and veterinary research. However, current approaches still heavily rely on manual examination under microscopes, even in modern clinical settings. This method, while standard, suffers from several limitations. Manual sperm detection and tracking are labor-intensive, time-consuming, and subject to variability between observers. Such inconsistencies can impact diagnostic accuracy and treatment outcomes, especially in high-throughput laboratory environments.

Moreover, existing Computer-Assisted Sperm Analysis (CASA) systems, although more objective, often depend on traditional image processing techniques such as thresholding, filtering, and edge detection. These methods may perform poorly in challenging conditions such as low-contrast images, high background noise, or overlapping sperm cells leading to false positives/negatives and unreliable tracking.

Another significant challenge is multi-species sperm classification and analysis, which is increasingly important in veterinary and comparative reproductive research. Each species exhibits distinct sperm morphology and motility patterns. However, current automated systems are often designed and trained for human sperm only, limiting their generalizability across species. This gap necessitates the development of intelligent, scalable models capable of handling inter-species variability in sperm structure and movement.

Furthermore, current CASA platforms generally lack integration with advanced deep learning models that could enhance performance. Modern neural networks, particularly convolutional and transformer-based models, offer robust capabilities for detecting and classifying visual patterns in biomedical imagery. Yet, their application in multi-species sperm tracking and classification remains underexplored.

This thesis addresses the need for a robust, deep learning-based framework capable of automating sperm detection, tracking, and classification across multiple species using microscopy images. The aim is to improve the accuracy, consistency, and efficiency of sperm analysis while making the approach adaptable and accessible for both clinical and research purposes.

4.2 Model selection

To build an efficient and scalable system for sperm analysis across multiple species, we separated our model design into two main components: sperm detection and tracking, and sperm species classification. Different architectures were chosen to best address the specific challenges associated with each task.

4.2.1 Detection and Tracking

In automating sperm detection and tracking, selecting an appropriate object detection model is crucial. Various models have been evaluated based on their performance metrics, especially in detecting small and densely packed objects, which is characteristic of sperm cells in microscopy images.

Comparative Analysis of Object Detection Models:

Faster R-CNN is renowned for its high accuracy, especially in detecting small and overlapping objects. However, its relatively slower inference speed renders it less suitable for real-time applications. For instance, Faster R-CNN has been reported to achieve an inference time of approximately 0.2 seconds per image, which is significantly slower compared to single-stage detectors like YOLO[40].

RetinaNet introduces a focal loss function to address class imbalance, enhancing detection performance on challenging datasets. While it offers a balance between speed and accuracy, it still falls short in real-time processing compared to newer models. The focal loss in RetinaNet effectively mitigates the issue of class imbalance by focusing training on hard examples [41].

Single Shot MultiBox Detector (SSD) is another single-stage detector that balances speed and accuracy. SSD is faster than two-stage detectors like Faster R-CNN but may struggle with detecting small objects due to its fixed-size default boxes [42].

EfficientDet is a family of models that scale efficiently with a compound scaling method. It achieves a good balance between accuracy and model size, making it suitable for deployment on devices with limited resources. However, its performance on small object detection tasks, such as sperm detection, may not be optimal [43].

YOLO (You Only Look Once) models have gained prominence due to their real-time detection capabilities.They have been optimized for real-time performance, and improved detection of small-scale targets.

The YOLO framework has undergone several iterations, each introducing enhancements to improve accuracy, speed, and adaptability.

Evolution of YOLO Architectures :

A. YOLOv2 (2016) - YOLO9000

YOLOv2 introduced significant improvements in training stability and bounding box prediction accuracy. The integration of Batch Normalization helped stabilize and accelerate the training process, while the adoption of Anchor Boxes aligned the model with techniques used in Faster R-CNN, enabling more effective bounding box predictions. Moreover, YOLO9000 extended the model's capabilities to recognize over 9,000 object categories by combining supervised and weakly supervised datasets [44].

B. YOLOv3 (2018)

YOLOv3 enhanced object detection performance across various object sizes by integrating a Feature Pyramid Network (FPN), allowing for more accurate multi-scale detection. The model also introduced the Darknet-53 backbone, a deeper and more robust feature extractor, which improved both classification and detection accuracy [44].

C. YOLOv4 (2020)

YOLOv4 focused on performance optimization by integrating Cross-Stage Partial (CSP) connections to reduce computational complexity without sacrificing accuracy. It also introduced the concepts of Bag of Freebies (BoF) and Bag of Specials (BoS), which provided performance enhancements during training and inference without additional computational cost [44].

D. YOLOv5 (2020)

YOLOv5 marked a shift by being implemented in PyTorch, which facilitated easier adoption and customization by the research community. It emphasized deployment by providing pre-trained models and a user-friendly interface, streamlining the application of object detection in real-world scenarios [44].

E. YOLOv6 (2022)

YOLOv6 brought a set of innovations tailored for efficiency and deployment. It employed RepVGG-inspired reparameterization techniques to improve speed and accuracy, especially for lightweight networks. Self-distillation was used across classification and regression tasks to enhance performance without increasing inference costs. Additional techniques such as Task-Aligned Learning (TAL), quantization optimization with RepOptimizer, and channel-wise distillation addressed the challenges of label assignment and quantization. An anchor-free design simplified decoding and improved generalization, while the architecture was tailored for industrial applications, especially on low-power GPUs like the Tesla T4 [44].

F. YOLOv7 (2022):

YOLOv7 introduced a range of innovations aimed at maximizing detection accuracy while maintaining real-time performance. The Trainable Bag-of-Freebies framework integrated trainable techniques such as re-parameterization and dynamic label assignment. The Extended-ELAN (E-ELAN) architecture used group convolutions to expand learning capacity without disrupting gradient flow. A planned re-parameterization strategy improved residual-based networks, while coarse-to-fine label assignment boosted detection precision. Additionally, a compound scaling method enabled efficient scaling of concatenation-based models. YOLOv7 achieved state-of-the-art real-time performance, surpassing both transformer-based and CNN-based detectors [44].

G. YOLOv8 (2023):

YOLOv8 continued the trend of superior real-time detection by achieving the highest accuracy (mAP) and speed (FPS) compared to previous YOLO models and other state-of-the-art detectors like Faster R-CNN and DETR. It introduced a unified framework that supported multiple tasks, such as object detection, instance segmentation, and classification. With an improved architecture featuring enhanced backbone and neck designs, YOLOv8 offered better feature extraction and fusion. The model adopted an anchor-free detection head and incorporated dynamic label assignment, advanced data augmentation, and optimized training strategies. It also provided an accessible API, a variety of pre-trained models (YOLOv8n to YOLOv8x), and compatibility with deployment frameworks such as TensorRT, ONNX, CoreML, and OpenVINO. Maintained by Ultralytics, YOLOv8 is open-source and community-driven [44].

H. YOLO-NAS (2023):

YOLO-NAS introduced the use of Neural Architecture Search (NAS) to automatically design an optimized model architecture, enhancing real-time detection performance. It featured a quantization-friendly design that maintained high performance with minimal loss when quantized, making it ideal for deployment on a range of hardware platforms. YOLO-NAS surpassed earlier YOLO versions in terms of mean Average Precision (mAP) and latency, striking a balance between accuracy and computational efficiency. It was released in three variants Small, Medium, and Large each optimized for different hardware and application constraints [44].

I. YOLOv9 (2024):

YOLOv9 introduced several advanced features to boost detection accuracy and efficiency. It implemented Programmable Gradient Information (PGI), an auxiliary reversible branch designed to preserve essential data across layers, improving gradient generation and minimizing information loss. The model leveraged the Generalized Efficient Layer Aggregation Network (GELAN), which combined the strengths of CSPNet and ELAN to maximize parameter efficiency and speed. YOLOv9 came in multiple variants YOLOv9t, YOLOv9s, YOLOv9m, YOLOv9c, and YOLOv9e catering to diverse needs in terms of speed and computational demands. It delivered superior mAP with fewer parameters and lower computational requirements, making it suitable for a variety of real-time tasks, including object detection and instance segmentation. The model also featured streamlined training procedures and broad compatibility with deployment frameworks like TensorRT and ONNX [45].

J. YOLOv10 (2024):

YOLOv10 introduces a groundbreaking innovation by eliminating the traditional Non-Maximum Suppression (NMS) step. This streamlines the object detection pipeline, reducing computational overhead and enhancing real-time performance. The model comes in six distinct variants YOLOv10-N, YOLOv10-S, YOLOv10-M, YOLOv10-B, YOLOv10-L, and YOLOv10-X each designed to balance latency and accuracy depending on specific deployment needs. With a mean Average Precision (mAP) reaching up to 54.4% and reduced latency, YOLOv10 demonstrates a significant leap forward in object detection efficiency and performance compared to its predecessors [46].

K. YOLOv11 (2024):

YOLOv11 brings improvements in detection accuracy through advanced attention mechanisms. It introduces the C2PSA (Convolutional Parallel Spatial Attention) block, which helps the model focus more effectively on critical regions of the image, especially in scenarios with occluded objects. Architecturally, it replaces older C2f blocks with optimized C3k2 modules that utilize smaller kernels (k=2), delivering faster computation without sacrificing feature extraction quality. YOLOv11 achieves higher mAP scores than YOLOv8 while using 22% fewer parameters in comparable variants, such as YOLOv11m versus YOLOv8m. Additionally, it expands capabilities to multi-task scenarios like oriented bounding box (OBB) detection and pixel-level instance segmentation [47].

L. YOLOv12 (2025):

YOLOv12 builds on architectural and computational efficiency with its Segmented Attention Mechanism. This system partitions the spatial domain of feature maps into distinct regions, allowing for more targeted attention while integrating FlashAttention to significantly reduce memory bandwidth demands. The model retains broad receptive fields with minimal overhead. Its backbone incorporates a Residual Efficient Layer Aggregation Network (R-ELAN), designed to improve gradient flow and promote feature reuse across different network depths. Standard convolutions are replaced with 7×7 separable convolutions to preserve spatial context and lower the parameter count, eliminating the need for explicit positional encoding [48].

M. YOLO-World (2025):

YOLO-World marks a shift toward open-vocabulary object detection, allowing the model to recognize objects outside of predefined categories in real time, without retraining. Operating at 52 FPS on a V100 GPU while maintaining a 35.4 AP on the LVIS benchmark, it leverages a CLIP-based text encoder to seamlessly fuse visual and linguistic features. This empowers YOLO-World to dynamically interpret user-defined prompts such as categories or descriptive phrases. Its zero-shot capabilities stem from pre-training on diverse datasets like Objects365 and GQA, alongside pseudo-labeled image-text pairs, enabling detection of unseen objects across over 1,200 LVIS categories. The prompt-then-detect paradigm facilitates deployment on edge devices by re-parameterizing vocabulary embeddings into model weights, making it ideal for domains such as autonomous driving, surveillance, and interactive AI, where adaptability to new object classes is critical [49].

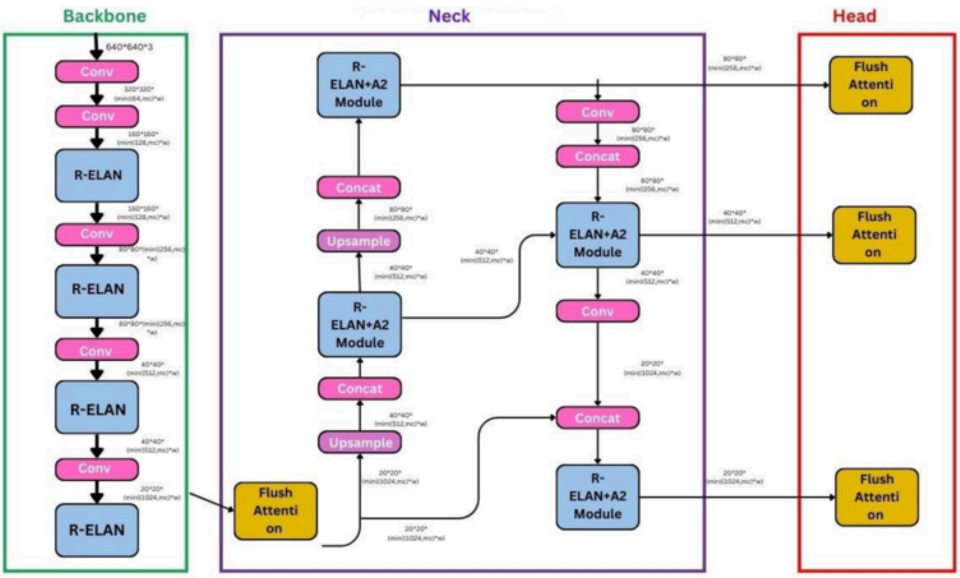


Figure 17: YOLOv12 architecture [48]

Benchmarking YOLO :

In this part, the benchmarking of Yolo was performed in detection on COCO dataset Explanation of Metrics:

- **mAPval 50-95 (Mean Average Precision at IoU 50-95):** A key metric for object detection, measuring precision-recall performance at different Intersection over Union (IoU) thresholds [57].
    - mAP 50-95 is the averaged mAP over IoU thresholds from 0.50 to 0.95 (with steps of 0.05), providing a comprehensive evaluation of model accuracy [57].
    - Higher values indicate better detection performance [57].

- **Speed CPU ONNX (ms):** Inference speed (latency) of the model when running on a CPU using the ONNX format. Measured in milliseconds (ms) lower values mean faster processing [57].
- **Speed A100 TensorRT (ms):** Inference speed when running the model on an NVIDIA A100 GPU optimized with TensorRT (a framework for high-performance deep learning inference). Lower values indicate faster processing, critical for real-time applications [57].
- **Params (M) (Model Parameters in Millions):** The total number of trainable parameters in the model. More parameters typically mean better accuracy but increased computational cost [57].
- **FLOPs (B) (Floating Point Operations in Billions):** Measures the total number of floating-point operations required for a single forward pass. A higher FLOP count means more computation is needed, which impacts efficiency and speed [57].

1. **Yolov8:**

Model	Size (pixels)	mAP_{val} 50-95	Speed CPU ONNX(ms)	Speed A100 TensorRT (ms)	Params(M)	FLOPs(B)
YOLOv8n	640	37.3	80.4	0.99	3.2	8.7
YOLOv8s	640	44.9	128.4	1.2	11.2	28.6
YOLOv8m	640	50.2	234.7	1.83	25.9	78.9
YOLOv8l	640	52.9	375.2	2.39	43.7	165.2
YOLOv8x	640	59.9	479.1	3.53	68.2	257.8

Table 03: Performance comparison of YOLOv8 models.[50]

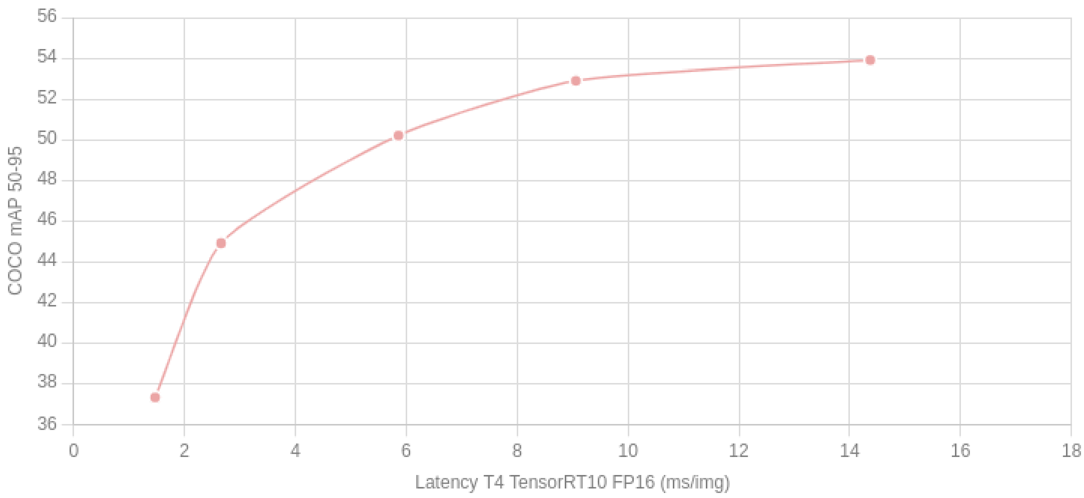


Figure 18: Speed vs Accuracy comparison of YOLOv8 model.[50]

2. Yolov-NAS

Model	Precision	mAP_{val} 50-95	Latency(BS=1 ms)	Params (M)
YOLO-NAS S	FP16 INT-8	47.5 47.0(-0.74)	3.21 2.36 (+0.85)	19.0
YOLO-NAS M	FP16 INT-8	51.55 51 (-0.55)	5.85 3.78 (+2.07)	51.1
YOLO-NAS L	FP16 INT-8	52.22 52.1(-0.12)	7.87 4.78 (+3.09)	66.9

Table 04: Results on NVIDIA's Tesla T4 GPU for YOLO-NAS models.[50]

3. Yolov9

Model	Size (pixels)	mAP_{val} 50-95	mAP_{val} 50	Params (M)	FLOPs(B)
YOLOv9t	640	38.3	80.53.1	2	7.7
YOLOv9s	640	46.8	63.4	7.2	26.7
YOLOv9m	640	51.4	68.1	20.1	76.8
YOLOv9c	640	53.0	70.2	25.5	102.8
YOLOv9e	640	55.6	72.8	58.1	192.5

Table 05: Performance comparison of YOLOv8 models.[50]

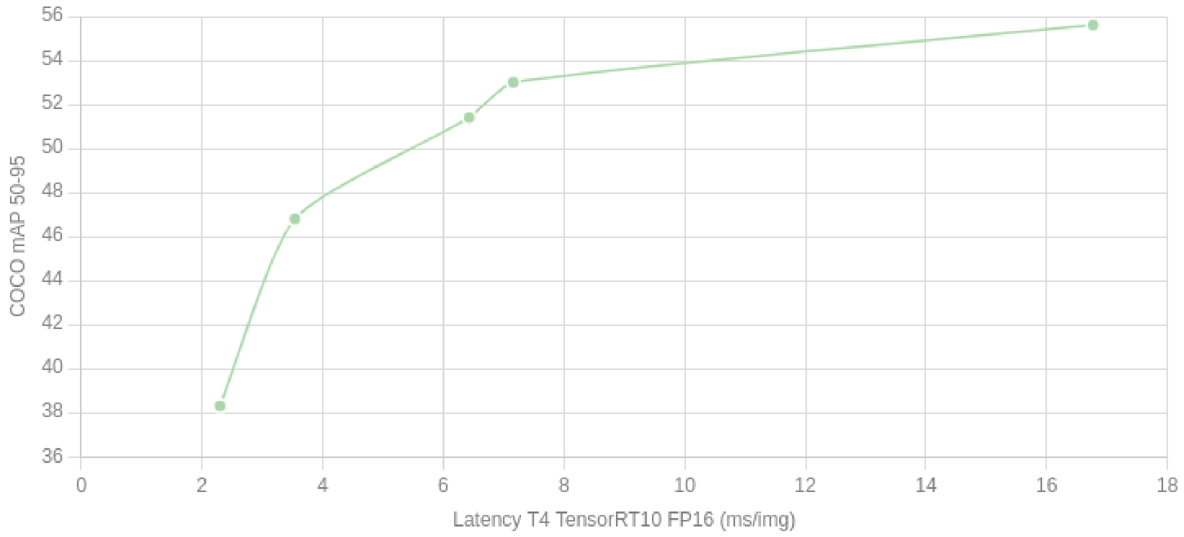


Figure 19 :Speed vs Accuracy comparison of YOLOv9 model.[50]

4. Yolov10

Model	**Input Size**	AP_{val}	**FLOPs (G)**	**Latency (ms)**
YOLOv10n	640	38.5	6.7	1.84
YOLOv10s	640	46.3	21.6	2.49
YOLOv10m	640	51.1	59.1	4.74
YOLOv10b	640	52.5	92.0	5.74
YOLOv10l	640	53.2	120.3	7.28
YOLOv10x	640	54.4	160.4	10.7

Table 06 : Performance comparison of YOLOv10 models.[50]

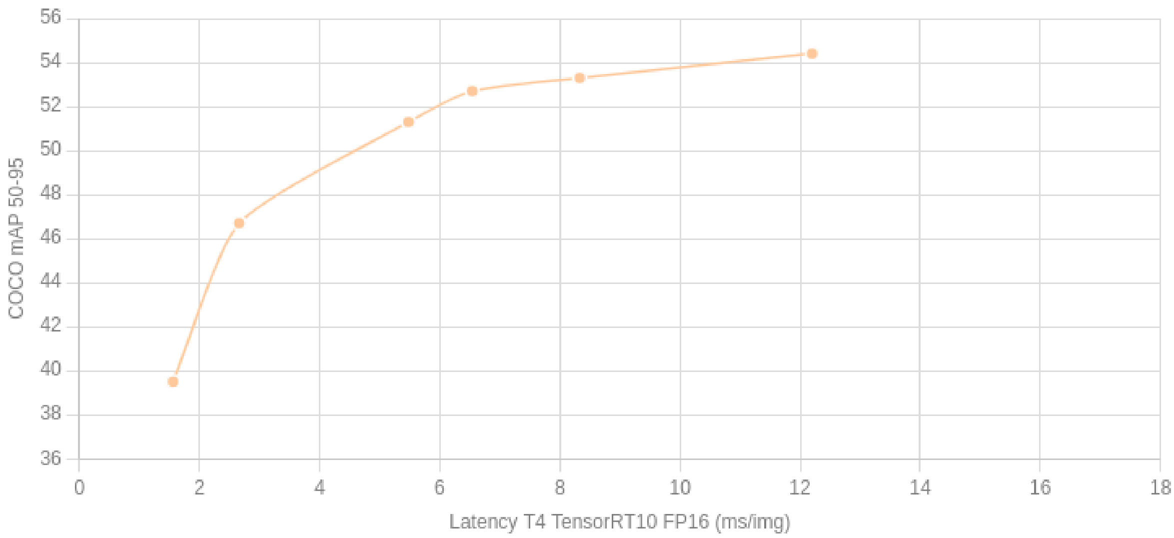


Figure 20 : Speed vs Accuracy comparison of YOLOv10 model.[50]

5. Yolov11

Model	Size	mAP^{val} 50-95	Speed (ms)		Params(M)	FLOP(B)
			CPU ONNX	T4 tensorRT		
YOLOv11n	640	39.5	56.1±0.8	1.5±0.0	2.6	6.5
YOLOv11s	640	47.0	90.0±1.2	2.5±0.0	9.4	21.5
YOLOv11m	640	51.5	183.2±2.0	4.7±0.1	20.1	68.0
YOLOv11l	640	53.4	238.6±1.4	6.2±0.1	25.3	86.9
YOLOv11x	640	54.7	462.8±6.7	11.3±0.2	56.9	194.9

Table 07: Performance comparison of YOLOv11 models. [50]

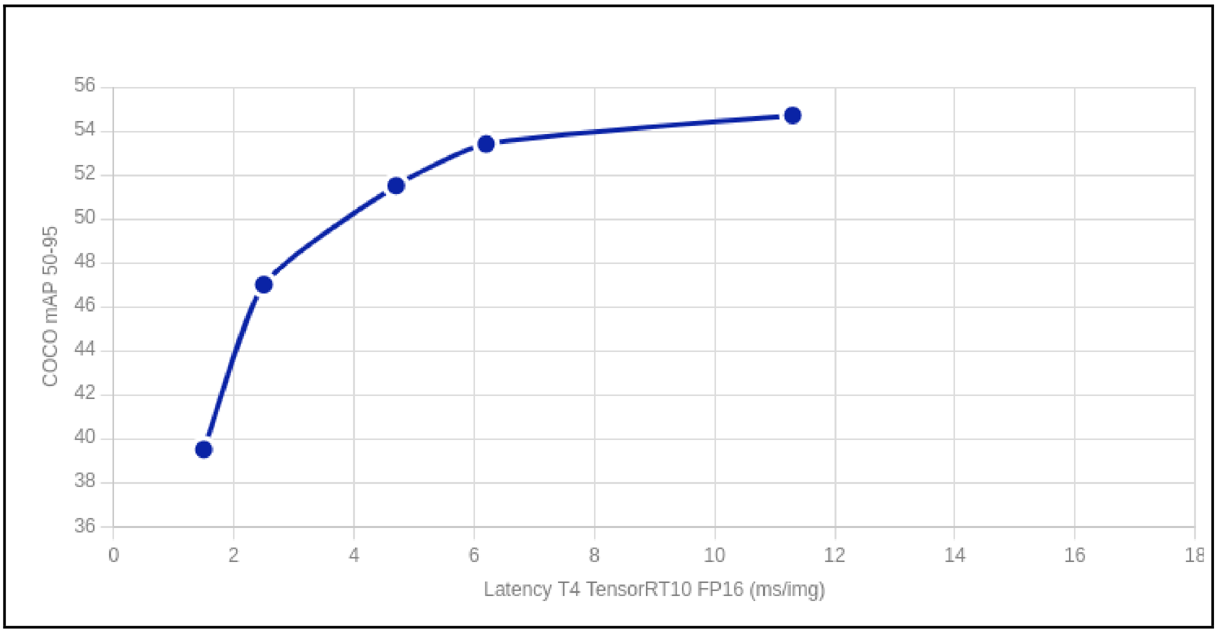


Figure 21 : Speed vs Accuracy comparison of YOLOv11 model [50].

6. Yolov-world

Model	mAP	mAP50	mAP75
yolov8s-world	37.4	52.0	40.6
yolov8s-worldv2	37.7	52.2	41.0
yolov8m-world	42.0	57.0	45.6
yolov8m-worldv2	43.0	58.4	46.8
yolov8l-world	45.7	61.3	49.8
yolov8l-worldv2	45.8	61.3	49.8
yolov8x-world	47.0	63.0	51.2
yolov8x-worldv2	47.1	62.8	51.4

Table 08 : Performance comparison of YOLOv8-world and YOLOv8-worldv2 models [50]

4.2.2 Model comparison:

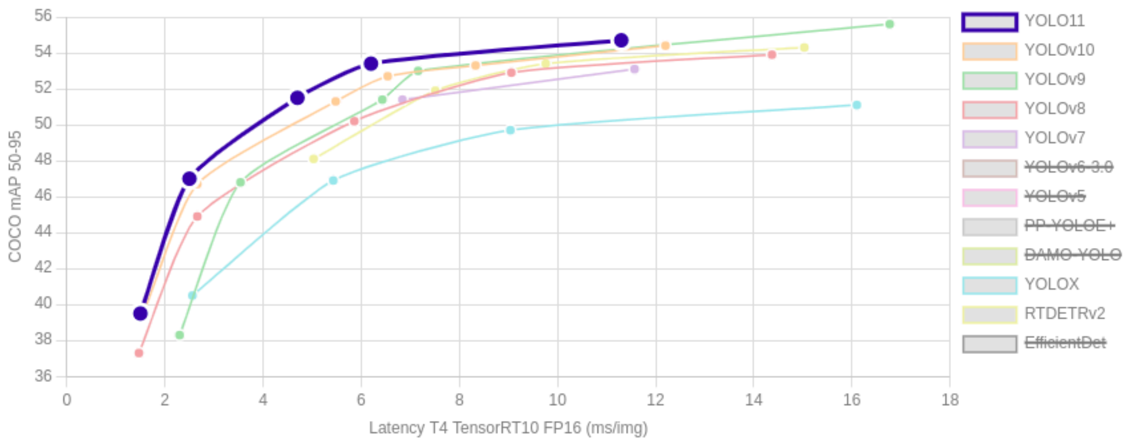


Figure 22 : Latency comparison of various object detection models on T4 TensorRT10 FP16 (ms/mg).[50]

The figure compares the latency of several state-of-the-art object detection models, including YOLOv11, YOLOv10, YOLOv9, YOLOv8, YOLOv7,YOLOX, and RTDETRv2. Latency is measured in milliseconds per megapixel(ms/mg) on a T4 GPU using TensorRT10 with FP16 precision. Lower latency values indicate faster inference times, which are critical for real-time applications.

we can observe the following:

The most recent addition, YOLOv11, is expected to outperform its predecessors by delivering reduced latency, making it more efficient for real-time processing. YOLOv10 also demonstrates competitive performance, likely striking a balance between rapid inference and reliable accuracy. Earlier versions like YOLOv9, YOLOv8, and YOLOv7 chart the model's evolutionary path, where each iteration aimed to lower latency while improving detection precision.

In parallel, YOLOX emerged as a high-performance alternative, engineered specifically for real-time scenarios. Its architecture allows for faster processing speeds compared to traditional YOLO variants, making it a strong candidate for applications requiring rapid decision-making.

Another noteworthy model, RTDETRv2, is tailored for real-time object detection and is optimized for speed. Its latency benchmarks are expected to be on par with or even surpass those of the YOLO series. Broadly speaking, the overall trend shows that newer architectures like YOLOv11 and YOLOv10 tend to achieve lower latency than their older counterparts, thanks to ongoing advancements in model design and optimization strategies.

The figure provides a clear comparison of the latency performance of various object detection models. We can use this information to select the most suitable model for a specific application, balancing latency, accuracy, and computational efficiency. Newer models like YOLOv11 and YOLOv10 are likely to offer the best performance for real-time applications, while older models may still be viable depending on the specific requirements.

Comparing Yolo NAS , Yolov8 and Yolov11:

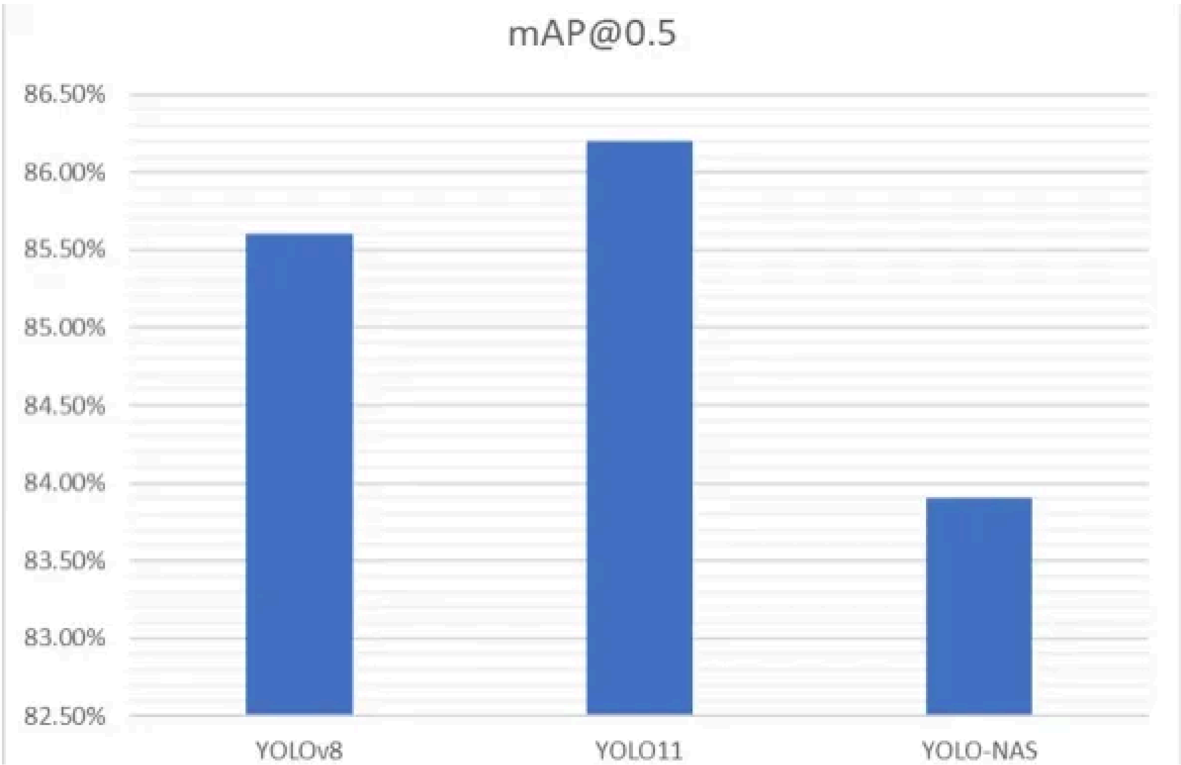


Figure 23 : comparing the performance of YOLO-NAS , yolov11, yolov8 trained on a custom vehicle dataset.[51]

Based on the evaluation results, YOLO-NAS shows a lower mAP@0.5 compared to YOLOv8 and YOLO11. This suggests that for this specific dataset, YOLO11 achieves the best accuracy, followed by YOLOv8, while YOLO-NAS lags behind. The dataset likely plays a crucial role in determining the effectiveness of YOLO-NAS. It is evident that YOLO-NAS does not always outperform other YOLO versions.

Comparing the performance of YOLO11 and YOLO-NAS as shown in Table 3 YOLO11 demonstrates better accuracy and overall performance . While YOLO-NAS has shown promise in other scenarios, it may not always be the optimal choice, especially when compared to YOLO11 and YOLOv8.

Comparing yolov11 and yolo-world:

In this section, we present the results of our experiments with various versions of YOLOv11 and YOLO-World models on the Visem dataset. All models were trained for 50 epochs using a GPU setup of T4x2. The performance metrics, including mAP@50, mAP@50-95, Precision, Recall, F1-score, and training time, are summarized in Table [7]:

Model	**mAP@50**	**mAP@50-95**	**Precision**	**Recall**	**F1–score**	**Training Time (s)**
yolov11s	0.9931	0.8580	0.9880	0.9808	0.9844	1866.56
yolov8s-worldv2	0.9945	0.8746	0.9902	0.9858	0.9880	2190.70
yolov11m	0.9949	0.9162	0.9947	0.9909	0.9928	3532.95
yolov8m-worldv2	0.9943	0.8819	0.9906	0.9829	0.9867	3647.25
yolov11l	0.9941	0.8742	0.9904	0.9831	0.9868	4613.70
yolov8l-worldv2	0.9949	0.9199	0.9964	0.9909	0.9937	5619.70

Table 09: Performance comparison of YOLOv11 and YOLOv8-World models on the Visem dataset. Training was conducted for 50 epochs using a T4x2 GPU setup.[50]

The evaluation of the YOLOv11 and YOLOv8-World models reveals several important insights based on their performance metrics. In terms of mAP@50, all the models exhibit outstanding accuracy, with scores nearing perfection indicating minimal differences among them for basic detection performance. However, when we shift our attention to the more demanding mAP@50-95 metric, a clearer distinction emerges. The YOLOv8-World models consistently take the lead, with the yolov8l-worldv2 model standing out by achieving an impressive score of 0.9199.

Looking deeper into precision, recall, and F1-score, it's evident that all models are performing at a high level. Yet again, the YOLOv8-World variants edge slightly ahead of their YOLOv11 counterparts, suggesting they may be more reliable in challenging detection scenarios. On the other hand, training time offers a different story. The YOLOv11 series has a clear advantage in terms of speed. For instance, yolov11s.pt completes training in just under 1,867 seconds, whereas the more complex yolov8l-worldv2.pt requires over 5,600 seconds, more than triple the time.

When it comes to selecting the best model, it ultimately depends on the specific requirements of the task. For users prioritizing accuracy above all, yolov8l-worldv2 is the optimal choice, delivering top-tier results across all accuracy-related metrics. If a balance between performance and training efficiency is preferred, yolov11m proves to be a solid candidate, offering near-peak performance with a much more reasonable training time. And for scenarios where training speed is critical such as frequent retraining or rapid deployment yolov11s.pt shines by completing its training quickly while maintaining respectable performance levels.

Lastly, recommendations vary by use case. For real-time applications, lightweight models like yolov11s.pt or yolov8s-worldv2 are ideal due to their faster inference and lower computational demands. Conversely, high-accuracy tasks benefit most from models like yolov8l-worldv2.pt or yolov8m-worldv2, which offer the most precise detection capabilities, even if they require more resources.

If accuracy is the most important factor and training time is not a constraint, yolov11m is the best choice. However, if we need a balance between accuracy and training time, yolov11m is excellent too. For faster training and inference, yolov11s.pt is the most efficient option.

Approche used:

For the tracking component of the system, we adopted a hybrid approach by integrating YOLOv11 for sperm cell detection with DeepSORT (Simple Online and Realtime Tracking with a Deep Association Metric) for maintaining consistent tracking identities across frames. This combination leverages YOLOv11's high-speed, high-accuracy object detection with DeepSORT's capability to track multiple objects efficiently using appearance features and motion information.

DeepSORT extends the original SORT algorithm by incorporating a deep appearance descriptor, allowing the model to handle occlusions and re-identify sperm cells even after temporary disappearance from the field of view. It associates detections across time using the Kalman Filter for motion prediction and the Hungarian algorithm for data association, enhanced with cosine distance between embedded appearance features [54].

4.2.3 Classification:

For the classification of sperm images across multiple species, EfficientNet was selected as the core model architecture. EfficientNet introduces a compound scaling method that uniformly scales network depth, width, and resolution using a fixed set of scaling coefficients. This approach has been shown to outperform conventional CNNs like ResNet, VGG, and Inception in terms of both accuracy and computational efficiency [52].

Unlike traditional deep networks that require manual tuning of dimensions, EfficientNet automates the scaling process while maintaining a lightweight model size. This characteristic is crucial for sperm classification, where high precision is needed but computational resources may be limited. Its performance on the ImageNet benchmark, where it achieved state-of-the-art results with significantly fewer parameters and FLOPs, supports its suitability for biological image classification tasks [52].

Moreover, EfficientNet's ability to generalize well across small and imbalanced datasets common in biomedical domains makes it particularly effective for multi-species sperm classification where annotated datasets are often scarce or noisy [53].

4.3 Feature engineering and model inputs

This section delves into the critical step of feature engineering and the selection of model inputs for the sperm analysis pipeline. Effective feature design is essential for enabling deep learning models to accurately detect, track, and classify sperm cells.

4.3.1 Detection and Tracking

To enhance detection and tracking performance, we applied tailored pre-processing and feature engineering steps on the input data. First, raw frames were extracted from the input video and resized to a consistent resolution to ensure uniformity across training and inference phases. Additionally, we implemented a patch-based image segmentation strategy using the patchify library, which divides each frame into non-overlapping patches of 256×256 pixels. This approach helps preserve the spatial resolution of smaller spermatozoa and improves detection sensitivity in dense scenes.

Each patch was independently passed through the YOLOv11 model, with predictions mapped back to global frame coordinates. This strategy ensures that local features are retained while providing a scalable way to detect objects in high-resolution videos. Detected bounding boxes, class scores, and confidence values were then fed into the DeepSORT tracker, which uses both spatial to maintain identity across frames.

For training, the YOLOv11 model consumed input images of size 640×640 pixels, annotated with bounding boxes in YOLO format. The corresponding dataset was specified using a YAML configuration file, ensuring compatibility with the Ultralytics framework. The input annotations were curated to accurately reflect the position and class of spermatozoa across multiple frames. During training, standard data augmentation techniques such as random flipping and color jitter were implicitly applied to increase model robustness and prevent overfitting.

4.3.2 Classification

To classify sperm species from microscopic images, we employed a supervised deep learning approach using EfficientNet-B0, a powerful convolutional neural network architecture. The model was initialized with pretrained weights from ImageNet and fine-tuned on our sperm species dataset using PyTorch and the timm library.

The dataset was organized into training and validation folders. We used standard data preprocessing steps, including resizing images to 224×224 pixels and applying ImageNet normalization. The data was loaded using ImageFolder, and DataLoader was used for batching and shuffling.

The classification head of EfficientNet was modified to match the number of sperm species classes. The training process used the Adam optimizer with a learning rate of 1e-4 and cross-entropy loss as the criterion. We implemented early stopping based on validation accuracy, saving the best model. TensorBoard was integrated to monitor the training metrics.

Once trained, the model was exported using torch.save and later reloaded for inference. The prediction pipeline accepts an image path and returns the predicted class after applying the same transformations used during training.

4.4 Training Protocols and Hyperparameter Settings

This section outlines the strategies and configurations used to train the deep learning models within the semen analysis pipeline.Effective training protocols play a vital role in ensuring model generalization and stability. In addition, hyperparameters like learning rate, batch size, number of epochs, and regularization techniques are fine-tuned to optimize performance across detection, tracking, and classification tasks.

4.4.1 Detection and Tracking

The training of the detection model was conducted using the Ultralytics implementation of YOLOv11, a lightweight and accurate object detector suited for real-time sperm tracking tasks. We trained the model using a custom dataset formatted according to YOLO standards and referenced through a YAML configuration file. Training was executed on a CUDA-enabled GPU to leverage accelerated computing capabilities.

The training session spanned 500 epochs, which provided a balanced trade-off between convergence and training time. We used an input image size of 640×640 pixels, ensuring a good balance between detection resolution and computational cost. The batch size was set to 4, suitable for the GPU memory constraints on the available hardware. The model was trained using default YOLOv11 settings unless explicitly modified.

To improve generalization, YOLOv11 applies various data augmentation techniques during training, including random scaling, flipping, mosaic augmentation, and color jittering. These augmentations help the model learn robust features even in the presence of noise or occlusion.

The best-performing weights, determined based on validation performance, were saved to a specified output directory. These trained weights were later used during inference and integrated into the detection and tracking pipeline described previously.

Hyperparameters such as learning rate, momentum, and weight decay followed YOLOv11's default configuration, as empirical tuning was not required beyond adjusting the batch size and number of epochs. Training was initiated using the Ultralytics `model.train()` method, passing all relevant parameters including the dataset path, image size, and run name.

4.4.2 Classification

The classification model was trained to categorize sperm images by species, leveraging the EfficientNet-B0 architecture from the TIMM library, which balances high accuracy and computational efficiency. A transfer learning approach was used, initializing the model with ImageNet-pretrained weights and adapting the final layer to output class probabilities for the eight sperm species in the dataset.

All input images were resized to 224×224 pixels and normalized using ImageNet statistics. Data augmentation was not applied during training to maintain the integrity of species-specific features. The dataset was split into training and validation subsets using the standard folder structure expected by torchvision.datasets.ImageFolder.

Training was performed using the Adam optimizer with a learning rate of 1e-4 and a batch size of 32, suitable for our hardware constraints. Cross-entropy loss was used as the objective function. The training loop spanned a maximum of 100 epochs and included early stopping based on validation accuracy with a patience of 10 epochs to prevent overfitting. During each epoch, performance metrics and loss values were logged using TensorBoard for better interpretability and debugging.

The best-performing model weights, determined by the highest validation accuracy, were saved and later reloaded for final evaluation and inference. The final trained model was exported as **sperm_species_classifier.pth** and used in the downstream classification pipeline.

4.5 Multi-task learning vs. modular pipeline

In this work, we adopted a modular pipeline approach where detection, tracking, and classification tasks were handled independently using specialized models. While multi-task learning (MTL) can unify these tasks under a single architecture, the modular design offers greater flexibility, easier debugging, and the ability to independently optimize each component for its specific objective.

The modular pipeline used YOLOv11 for sperm detection and tracking, followed by an EfficientNet-B0 classifier for species identification. This separation allowed for task-specific data preprocessing, hyperparameter tuning, and model selection, which collectively enhanced performance and interpretability. For example, YOLOv11 benefited from image augmentations such as mosaic and color jittering to improve object localization, whereas the classifier relied on cleaner preprocessing to preserve species-defining features.

By contrast, a multi-task learning strategy would involve a shared backbone trained jointly on detection and classification objectives. While this approach can promote parameter sharing and improve generalization in some cases, it also introduces complexity in loss balancing, task interference, and data alignment. Moreover, in our application, the detection and classification tasks involve fundamentally different spatial and semantic representations, making full integration less practical without extensive architectural engineering.

Overall, the modular pipeline proved more efficient and manageable for our sperm analysis tasks, particularly in a research context where model interpretability, modular retraining, and performance isolation are key priorities.

4.6 Conclusion

This chapter detailed the comprehensive methodology behind developing an AI-driven system for sperm detection, tracking, and classification. It began with a clear formulation of the problem, outlining the specific objectives and constraints. The process of model selection was explored next, including tailored architectures for both detection and classification, and a comparative analysis to assess performance trade-offs. The chapter also covered the design of relevant input features and engineering strategies to optimize model learning. Training procedures were then specified, including hyperparameter configurations for each task. Finally, the discussion of multi-task learning versus modular pipelines highlighted key considerations in balancing efficiency, accuracy, and system complexity. Collectively, these methodological foundations form the backbone of the pipeline used for robust, species-aware sperm analysis.

Chapter 5: Experimental Results and Performance Analysis

This chapter presents a comprehensive evaluation of the experimental results obtained through the application of deep learning techniques for automated semen analysis. Our work aims to enhance the accuracy and efficiency of traditional semen assessment by addressing three critical tasks: detection, tracking, and classification of sperm cells (or sperm species, depending on context). These tasks are essential for assessing male fertility and have direct implications in both clinical diagnostics and reproductive research.

We begin by presenting the results of sperm cell detection, evaluating the model's ability to accurately localize sperm cells in microscopy images. This section highlights the robustness of our detection approach and its ability to distinguish between sperm cells and background noise or debris.

Next, we investigate the task of tracking sperm movement over time. Here, we assess the system's performance in maintaining identity across frames, estimating trajectories, and analyzing motility patterns. This tracking component is crucial for quantifying sperm dynamics, which play a key role in fertility evaluation.

Following this, we evaluate the classification module, which categorizes detected sperm cells based on species (e.g., human, bovine, murine).

Finally, we synthesize the findings from each task, analyze the strengths and limitations of the proposed system, and outline its practical implications. We conclude the chapter by summarizing our key contributions and suggesting avenues for future improvements and extensions, including the integration of real-time analysis or deployment in clinical/laboratory environments.

5.1 Evaluation Metrics

To thoroughly evaluate the system across its different tasks, we employed task-appropriate metrics for detection, tracking, and classification:

- **Accuracy:** Measures the proportion of total predictions that are correct [57].

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$$

- **Precision**:
  - **Classification :** Indicates the percentage of instances predicted as class *c* that actually belong to *c* [57].
  - **Detection:** Precision evaluates the proportion of predicted bounding boxes that correctly match a ground-truth object [55].

$$Precision = \frac{TP}{TP+FP}$$

- **Recall:**
  - **Classification :** Measures the proportion of instances of class c that were correctly predicted [57].
  - **Detection :**indicates how many ground-truth objects were successfully detected by the model [55].

$$Recall = \frac{TP}{TP+FN}$$

- **F1-Score :** Combines precision and recall into a single measure using the harmonic mean.[57]

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

- **mAP@0.5 (mean Average Precision at IoU ≥ 0.5) :** This score averages the precision values over all classes, considering predictions correct if their IoU with ground truth is ≥ 0.5 [56].
- **mAP@[0.5:0.95] :** This is the mean of AP scores at IoU thresholds from 0.5 to 0.95 in steps of 0.05. It offers a more stringent evaluation of object localization [56].
- **Intersection over Union (IoU) :** IoU quantifies the ratio between the area of overlap and the area of union of the predicted and actual bounding boxes.

$$IoU = \frac{Area\ of\ overlap}{Area\ of\ Union}$$

5.2 Training Results :

This section outlines the strategies and configurations used to train the deep learning models within the semen analysis pipeline. Effective training protocols such as data splitting, augmentation, loss function selection, and optimization routines play a vital role in ensuring model generalization and stability. In addition, hyperparameters like learning rate, batch size, number of epochs, and regularization techniques are fine-tuned to optimize performance across detection, tracking, and classification tasks.

5.2.1 Detection and tracking model results:

In this section, we present the training results of the detection and tracking model, highlighting the outcomes of two successive fine-tuning stages designed to enhance performance across diverse sperm analysis scenarios.

The first fine-tuning phase was conducted using the ViSEM dataset, a publicly available semen analysis dataset that offers a rich variety of sperm cell imagery under controlled conditions. This dataset allowed us to adapt our pre-trained model to the domain of sperm detection and tracking, leveraging its annotations to improve localization and temporal tracking accuracy. Following training, the model demonstrated strong performance metrics on the ViSEM validation set, including precise sperm head localization, robust frame-to-frame identity tracking, and consistency in motility path estimation.

To further adapt the model to our specific application and real-world microscopy conditions, a second fine-tuning phase was performed on our custom dataset, which originates from a real-world clinical study conducted in Algeria. as described earlier in this work.Fine-tuning on this data .

5.2.1.1 Fine-Tuning on the ViSEM Dataset:

- **Training Configuration**

Parameter	Value
epochs	300
batch size	8
patience	12
iou	0.05

lr0 (initial learning rate)	0.001
lrf	0.2
amp	True

Table 10 : Training Configuration *for sperm detection task*

- **Quantitative Results**

The training converged after 89 epochs, with early stopping patience set to 12 epochs. This indicates the model reached optimal performance without overfitting, as training halted once validation performance plateaued.

The final evaluation metrics reported are as follows:

Metrics	**Visem dataset result**
Precision (P)	0.991
Recall (R)	0.938
mAP@0.5	0.961
map@0.5:0.95	0.824

Table 11 : Quantitative Results for sperm detection task

The model demonstrated outstanding performance in object detection tasks, achieving a precision of 0.993, which means it almost never confused non-sperm regions with actual sperm cells. Its recall score of 0.936 highlights its effectiveness in identifying most of the true positives, showing strong sensitivity. Furthermore, the mean Average Precision at a 0.5 Intersection over Union (mAP@50) was recorded at 0.961, underscoring its accuracy in localizing sperm cells correctly. Even when evaluated across a broader range of IoU thresholds (mAP@0.50:0.95), the model maintained a robust performance with a score of 0.824, indicating its ability to handle various levels of detection difficulty with consistency.

These results suggest that the YOLO-based architecture fine-tuned on the ViSEM dataset is highly reliable and well-suited for sperm cell detection in standardized environments.

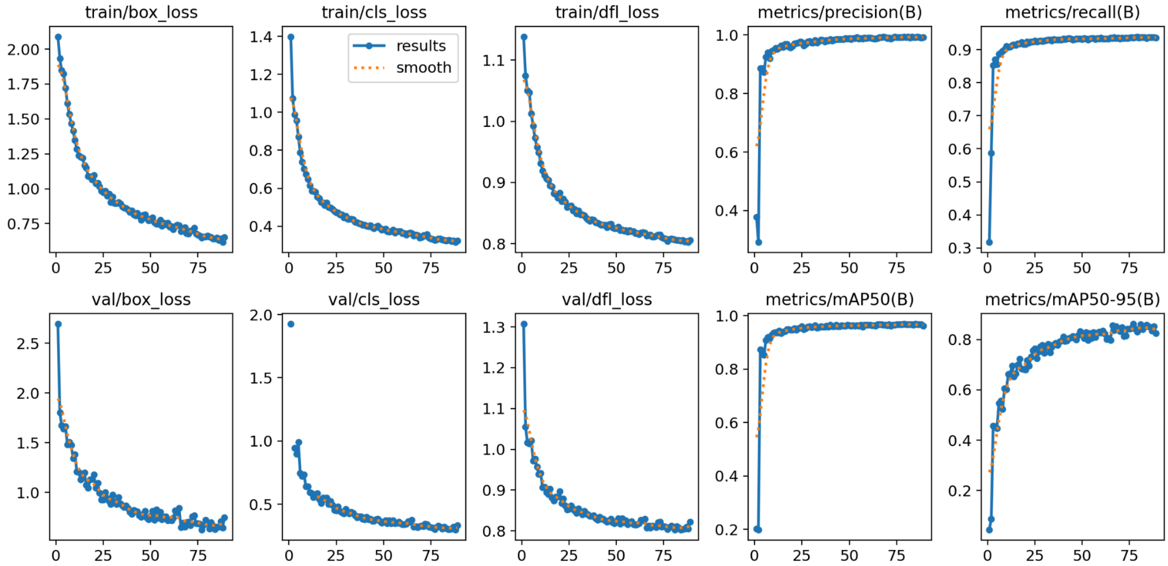


Figure 24 : Loss and metrics curves of train and val of Yolov11 on VISEM dataset

- **Results and visualisation:**

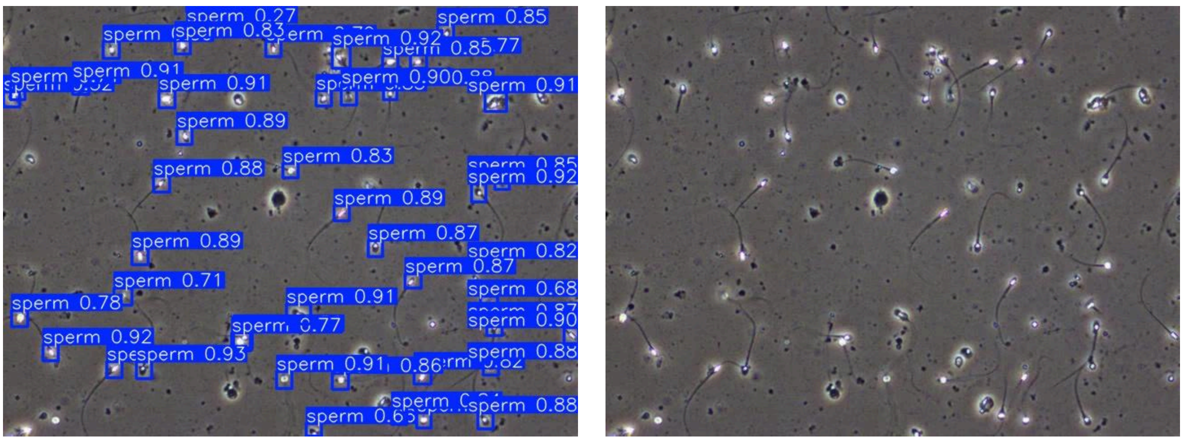


Figure 25: Example 1 of result of our model on visem dataset

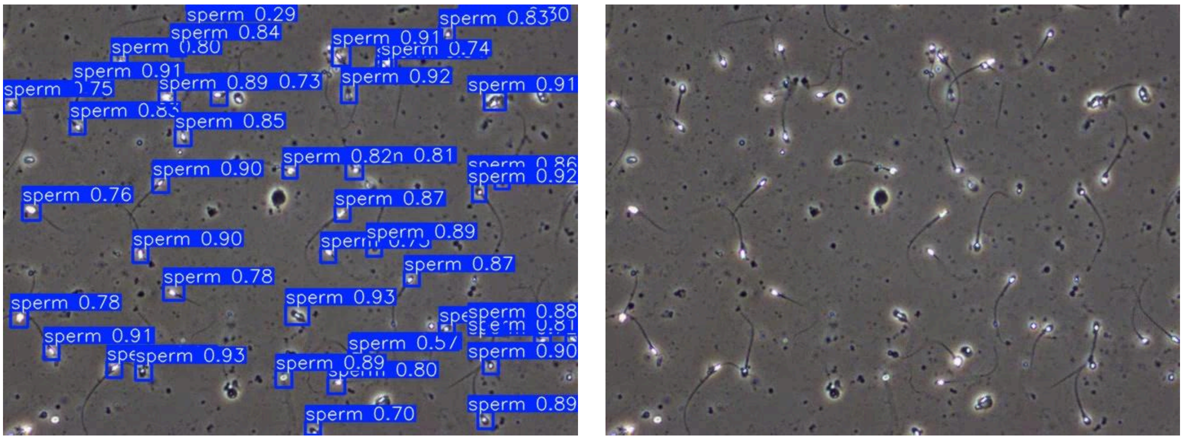


Figure 26 : Example 2 of result of our model on visem dataset

5.2.1.2 Fine-Tuning on Our Custom Dataset

- **Training Configuration**

The training configuration used for fine-tuning on our custom dataset was identical to that applied in the ViSEM dataset experiment. This includes:

epochs	300
batch size	8
patience	12
iou	0.05
lr0 (initial learning rate)	0.001
lrf	0.2
amp	True

Table 12 : Training configuration of sperm detection task

- **Quantitative Results**

The training converged after 141 epochs, with early stopping patience set to 12 epochs. This indicates the model reached optimal performance without overfitting, as training halted once validation performance plateaued.

The final evaluation metrics reported are as follows:

Metrics	**Custom dataset result**
Precision (P)	0.822
Recall (R)	0.829
mAP@0.5	0.874
map@0.5:0.95	0.523

Table 13 : results of detection sperm on our custom dataset

The model demonstrates strong detection capabilities on our real-world dataset, achieving a high precision of 82.2% and recall of 82.9%. The mAP@0.5 score of 87.4% indicates reliable localization performance, while the mAP@0.5:0.95 value of 52.3% reflects a reasonable ability to generalize across IoU thresholds. These results confirm the applicability of the approach in realistic clinical or research scenarios involving sperm detection and tracking .

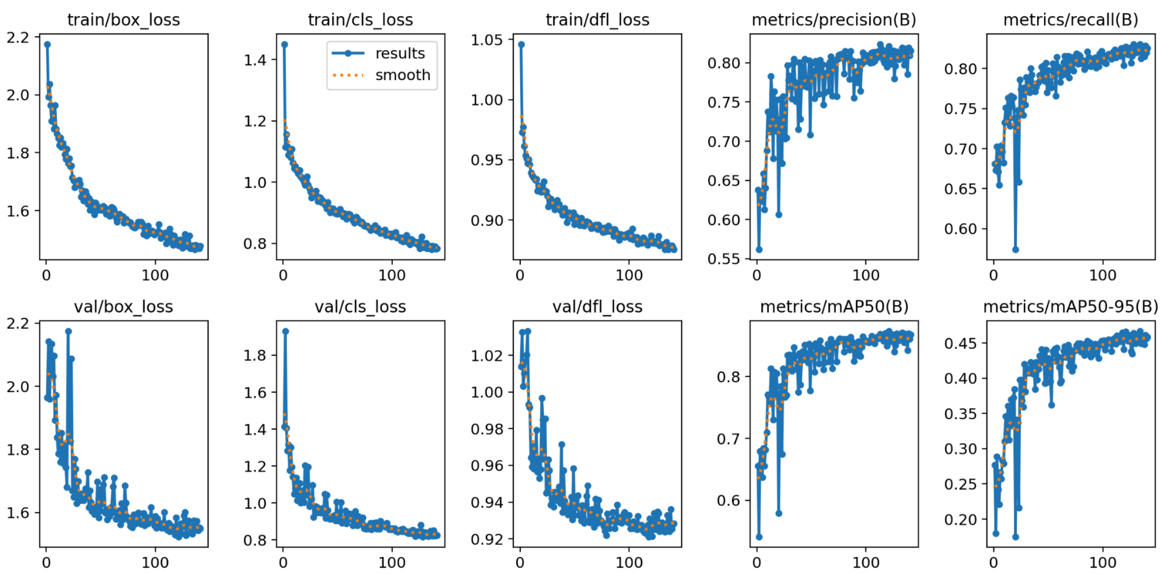


Figure 27 : Loss and metrics curves of train and val of Yolov11 on our custom dataset

- **Results and visualisation:**
  - **Detection:**

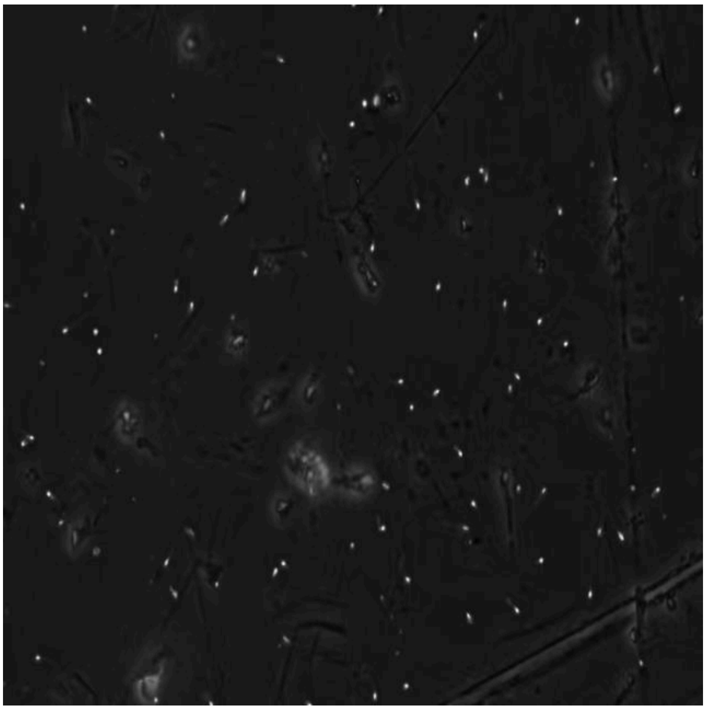


Figure 28:Original image before processing

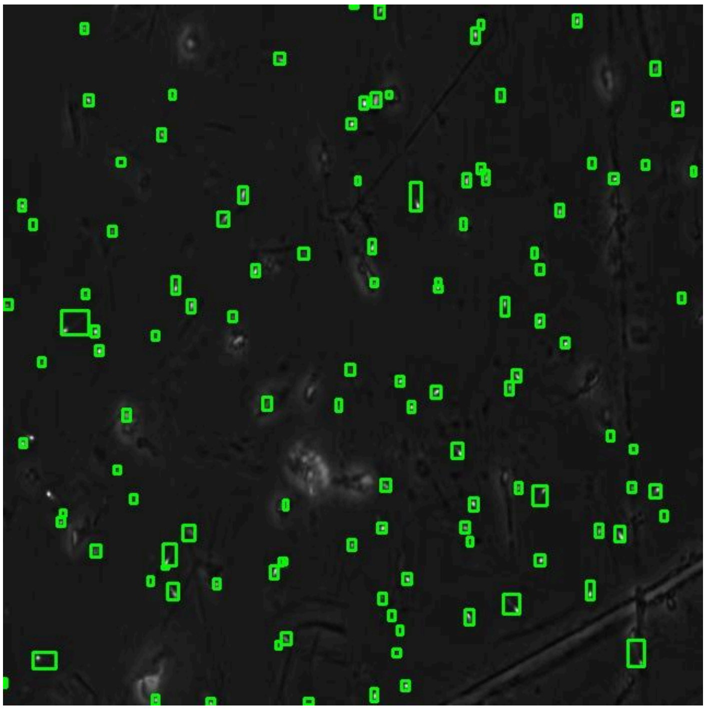


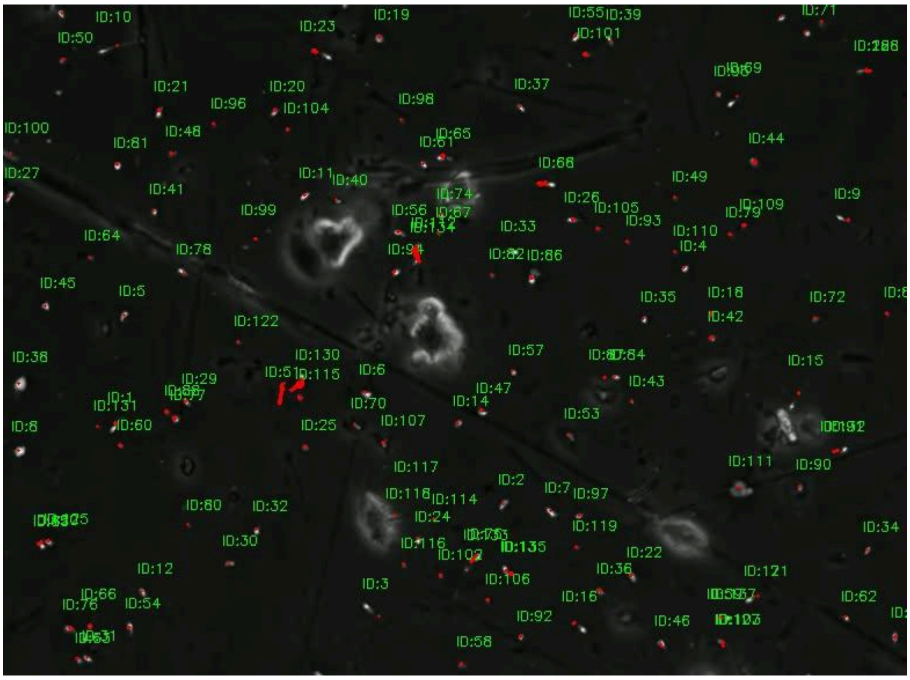
Figure 29 : Output Display of the Original Image

- **Tracking:**

To evaluate sperm cell motility and movement patterns, we employed the DeepSORT tracking algorithm on successive frames extracted from microscopic video sequences. DeepSORT enabled us to assign consistent identities to individual sperm cells across time, allowing precise tracking of their trajectories throughout the video. From these trajectories, we extracted a series of motion-based features, such as instantaneous velocity (VCL curvilinear velocity), straight-line velocity (VSL), average path velocity (VAP).

Once the features were computed for each detected and tracked spermatozoon, we aggregated the results across all detected instances to generate frame-level and sample-level statistics. This included averaging velocities to assess general motility trends and classifying individual sperm into categories like progressive, non-progressive, or immotile. These aggregated motion descriptors formed a critical part of the quantitative semen analysis.

This combination of object detection and multi-object tracking provided a robust, automated framework to mimic the traditional manual sperm tracking conducted by clinicians while enhancing precision, scalability, and reproducibility.



Average VCL: 0.41 µm/s Average VSL: 0.15 µm/s Average VAP: 0.41 µm/s

Figure 30: Last video frame showing sperm paths with velocity result count

5.2.2 Species semen classification model results:

In this section, we present the results of our semen species classification model, developed to identify and distinguish between different semen species from microscopic images. Unlike the detection and tracking module, which is based on object detection frameworks, this classification model leverages the EfficientNet architecture due to its balance between accuracy and computational efficiency.

The classification pipeline was trained separately and is tailored specifically for image-level predictions, aiming to categorize each input sample into its corresponding species class. We evaluated the model's performance on our custom dataset that contains images from 8 different species.

- **Training Configuration**

Parameter	Value
epochs	100
batch size	32
patience	10
optimizer	Adam
Initial learning rate (lr)	10^{-4}
input size	224×224
loss function	Cross-entropy
AMP (mixed precision)	True

Table 14 : Training configuration for the classification task

The model was trained using transfer learning with pretrained ImageNet weights. Early stopping was applied to monitor validation accuracy, with a patience of 10 epochs.The dataset was split into training and validation sets using a folder-based structure compatible with torchvision.datasets.ImageFolder

- **Quantitative Results:**

The model training converged after 21 epochs, triggered by early stopping once validation accuracy plateaued (patience = 10 epochs). The training demonstrated very high classification performance on both training and validation sets, suggesting excellent generalization.

Metric	Result
Accuracy	0.999
Precision	0.999
Recall	0.999
F1-Score	0.999

Table 15 : Quantitative results for classification task

These metrics confirm that the classifier performs robustly across all sperm species categories. The model effectively learned discriminative features, achieving perfect accuracy on the validation set while avoiding overfitting. This makes it well-suited for use in research pipelines where accurate species identification is essential.

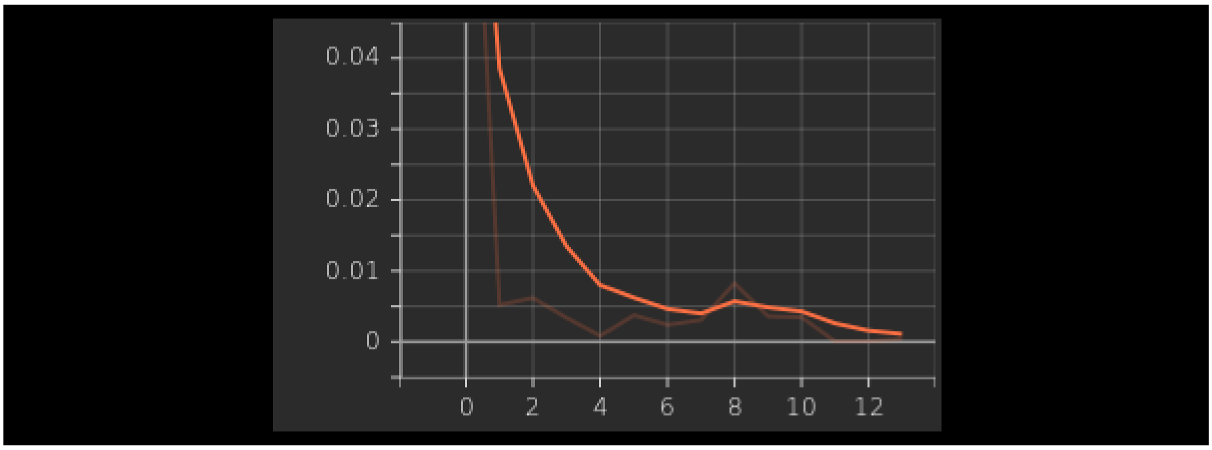


Figure 31 : Training and validation Loss Curve

The training loss plot reflects a **rapid decline** in loss within the first few epochs, reaching very low values and plateauing thereafter. By the final epochs, the loss value hovers near zero, which is typical for a highly confident classifier on well-learned training data.

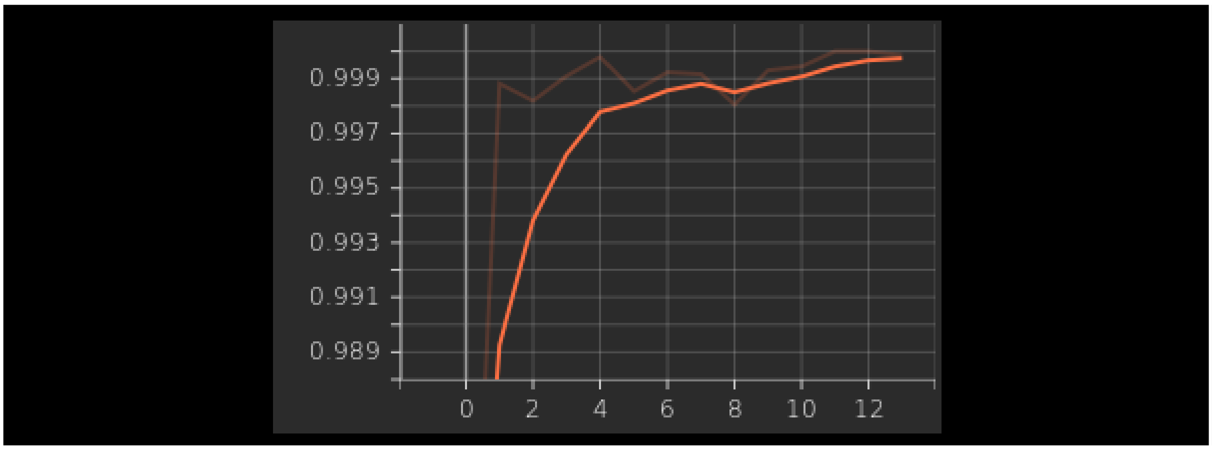


Figure 32 :Training and validation Accuracy Curve

The training accuracy curve shows a very steep increase within the first few epochs, quickly approaching near-perfect accuracy. By epoch 10, the model surpasses 99% accuracy and remains consistently high throughout the training process. This indicates that the model learned to classify the training data very effectively with minimal fluctuation, demonstrating strong fitting capacity and stable convergence.

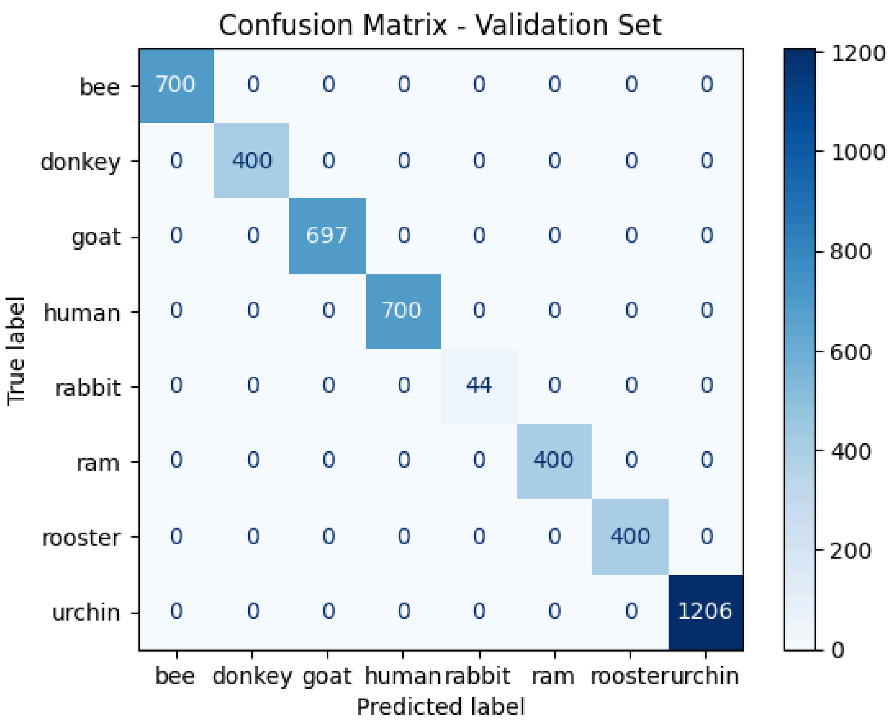


Figure 33: Confusion matrix on the validation set

The confusion matrix paints a remarkable picture of the model's classification performance on the validation set,one of absolute precision. Every single sample has been accurately classified, with no instances falling into incorrect categories. Each class aligns perfectly with its predicted counterpart, showing zero confusion across the board.

Notably, even the rabbit class despite having only 44 examples achieves flawless accuracy. This outcome strongly suggests that the model isn't skewed by class imbalance and maintains fairness across both dominant and minority classes.

Such results imply several things: the model has developed a strong ability to generalize to unseen data, it has successfully learned highly distinctive features for each category, and it likely benefited from a robustly curated dataset and carefully optimized training process.

5.3 Conclusion

This chapter demonstrated the effectiveness of our AI-driven pipeline through detailed performance evaluations. High precision, strong recall, and robust detection metrics confirmed the reliability of the sperm detection and tracking models. The classification results also showed excellent generalization across species, supported by consistent accuracy and well-structured confusion matrices. Together, these findings validate the system's potential for real-world deployment in veterinary and clinical environments.

Chapter 6: Real-World Deployment and Integration

This chapter explores the practical transition from research to application, focusing on how AI-based semen analysis systems can be integrated into real-world environments. It highlights the strategies for embedding these models into existing laboratory workflows, developing user-friendly interfaces for clinical and veterinary use, and envisioning future system architectures for broader deployment.

6.1 Integration into lab pipelines

Building upon the robust performance of our detection, tracking, and classification models, the project has entered a pivotal stage: transitioning from controlled research settings to real-world laboratory environments. While full-scale deployment is still underway, meaningful collaborations have been established with a private clinical laboratory and the Biology Department at Abderrahmane Mira University. These partnerships have played a vital role in facilitating system evaluation by providing access to laboratory equipment, anonymized sample data, and expert feedback key elements in shaping a deployment-ready solution.

The integration pipeline is designed to support real-time operation, enabling direct analysis of live video feeds from microscope cameras. Each incoming frame is instantly processed by the AI system, which detects, tracks, and classifies individual sperm cells by species. Results are presented through an interactive monitoring interface, with the option to export findings as structured clinical reports ensuring both immediate visualization and standardized documentation.

Preliminary validation using pre-recorded microscopy videos has demonstrated encouraging outcomes, with the system maintaining high detection accuracy and stable performance under realistic imaging conditions. Ongoing development efforts are focused on enhancing compatibility with diverse lab hardware, reducing latency through GPU acceleration, and refining the user interface based on technician feedback to ensure seamless integration into daily workflows.

This practical integration phase marks a critical milestone in validating the model’s real-world applicability and sets the foundation for a forthcoming pilot deployment within the collaborating laboratory environments.

6.2 Application Design and Features

The core objective of the application is to bridge the gap between AI-driven semen analysis research and its practical use in clinical and veterinary settings. Designed as a hybrid system, the solution combines a desktop-based analysis tool with a centralized cloud platform ,together providing seamless sperm evaluation, client management, and system maintenance.

The desktop application is tailored for lab technicians, enabling them to perform sperm detection, tracking, and species classification directly from microscopy videos. The interface supports live or recorded video inputs, real-time visualization of results, and structured report generation. Users can also manage client records, retrieve historical data, and request subscription renewals through an integrated user dashboard.

On the other side, the web-based platform provides administrators and engineers with a centralized hub for overseeing the system. Key features include the ability to create and manage clinic accounts, oversee billing and subscription statuses, and push updates to the underlying AI models. Engineers can also deploy new versions of the model or update the analytical pipeline directly through the platform.

By combining real-time analysis on the desktop with cloud-level control and scalability, this dual-structure design ensures both performance in laboratory environments and centralized management of the broader ecosystem.

From a technical perspective, the system was developed using a modular and scalable stack. The desktop application's interface was built using React and styled with Tailwind CSS, offering a responsive and clean user experience. Its local backend was implemented using Node.js, responsible for handling video inputs, user operations, and communication with the AI model service.

The AI inference service, which performs detection, tracking, and classification, was developed using FastAPI, allowing fast and lightweight interaction with the model. All user and analysis data are managed using MongoDB, chosen for its flexibility and ease of integration with the Node.js environment.

To ensure maintainability and portability, the desktop backend, AI service, and database were all containerized using Docker, allowing for smooth deployment and consistent environments across different systems.

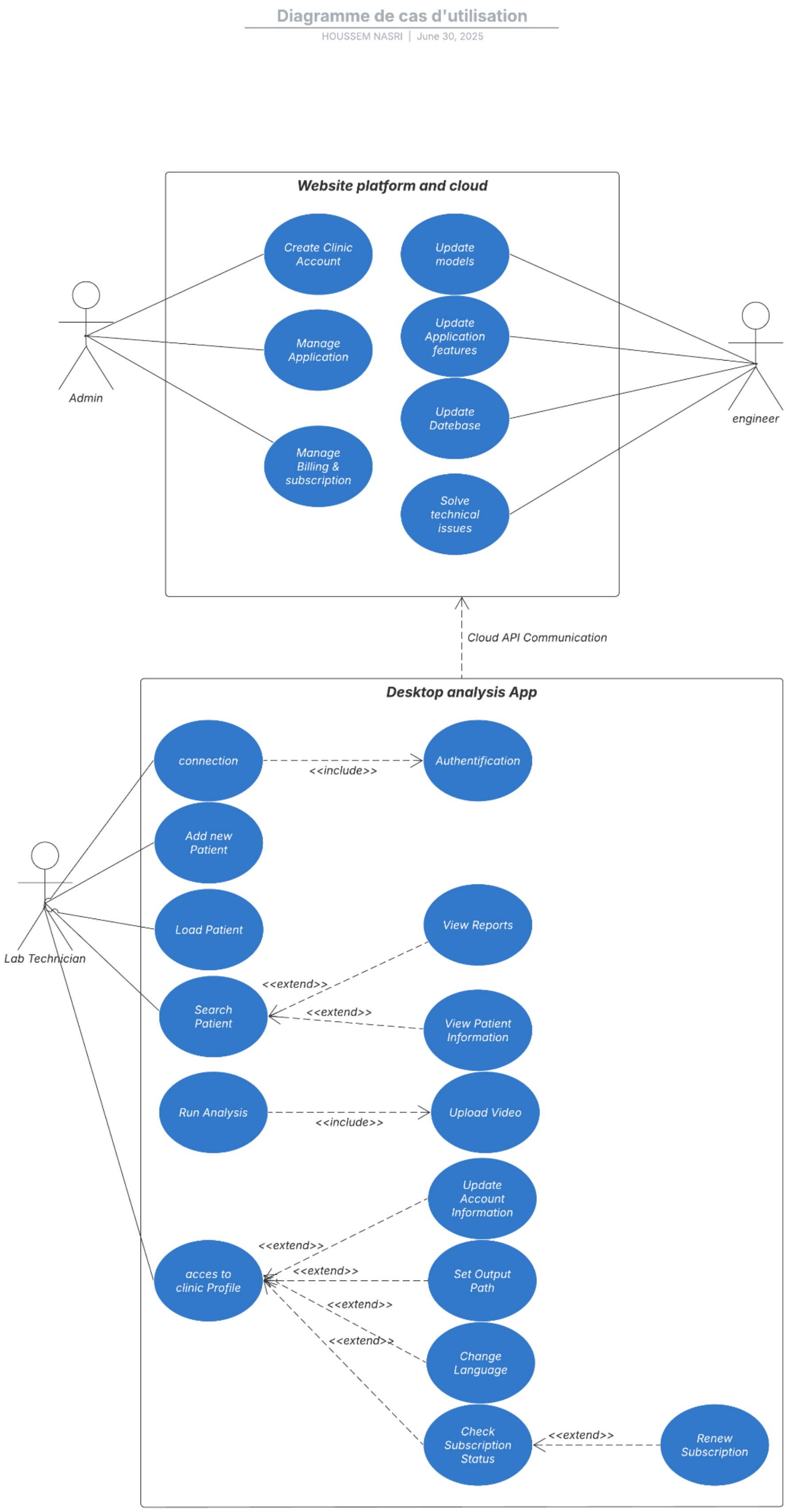


Figure 34: Use case diagram of our implementation

The application's data layer was implemented using MongoDB, a document-oriented NoSQL database. Its flexibility made it ideal for handling the diverse and nested structures of client profiles, analysis results, subscription plans, and model metadata.

Each client entry stores personal information, analysis history, uploaded video references, and generated reports, all structured in easily queryable JSON-like documents. Subscription data includes activation status, expiration timers, and renewal requests, ensuring accurate access control for the desktop app.

MongoDB was integrated via Mongoose (Node.js ORM) on the desktop backend and accessed via RESTful API endpoints exposed by FastAPI on the inference server. This unified design allowed both the local client and the cloud dashboard to interact with the same synchronized data source securely and efficiently.

Role-based access control (RBAC) was implemented to restrict sensitive database actions (like account creation or model updates) to authenticated admin or engineer users only.

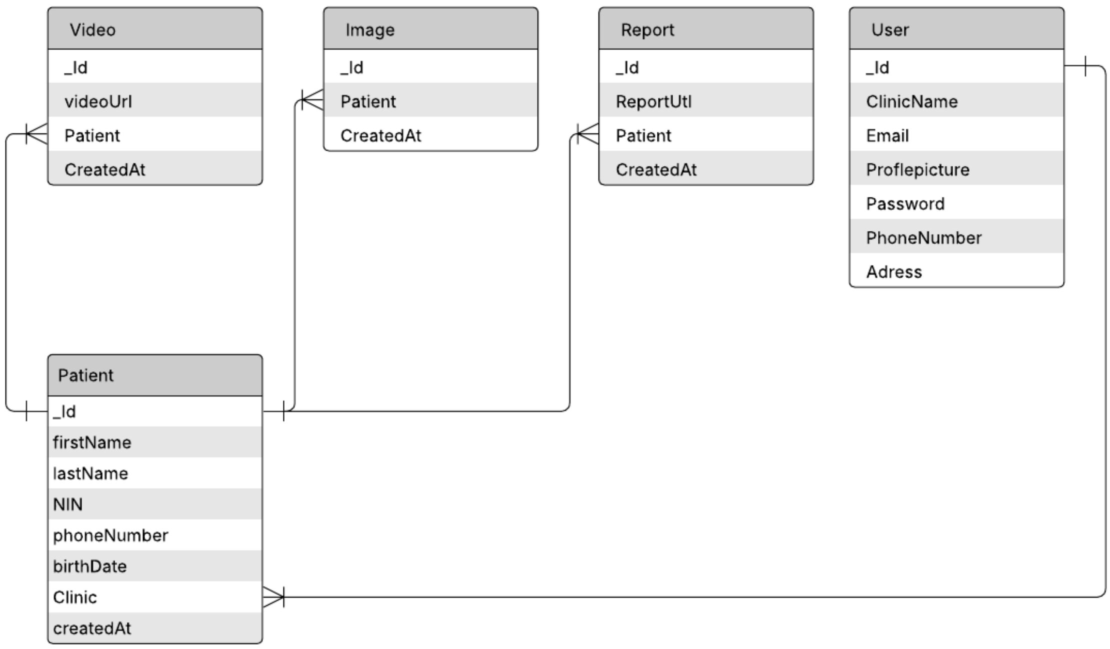


Figure 35: DataBase Schema

6.3 Interface for Clinical and Breeding Program Use

To ensure that the system is accessible and useful to its intended users,namely clinical labs and breeding program operators,a dedicated user interface has been developed with a strong focus on practicality, clarity, and workflow compatibility.

The desktop interface is specifically tailored to laboratory technicians who perform daily semen analyses. It offers a streamlined experience that guides the user through each step: uploading video samples or capturing live feeds, initiating analysis, reviewing real-time detection and classification overlays, and saving structured reports. Each client's data is stored locally and/or synced with cloud services when applicable, allowing quick retrieval of historical results.

For breeding programs, where tracking fertility performance and managing animal data is essential, the interface includes modules to associate semen analysis reports with individual animals. Users can review trends over time, compare results across collection dates, and export summaries to support reproductive decision-making.

Complementing the desktop tool, the web dashboard allows clinic managers and veterinary staff to oversee broader operations,managing user roles, reviewing billing cycles, handling subscription renewals, and monitoring system usage. It also supports admin alerts and update notifications to ensure clinics always operate with the most recent models and features.

Together, these interfaces aim to reduce friction in day-to-day workflows, improve the traceability of analysis outcomes, and bring AI-powered semen evaluation into the hands of professionals without requiring deep technical knowledge.

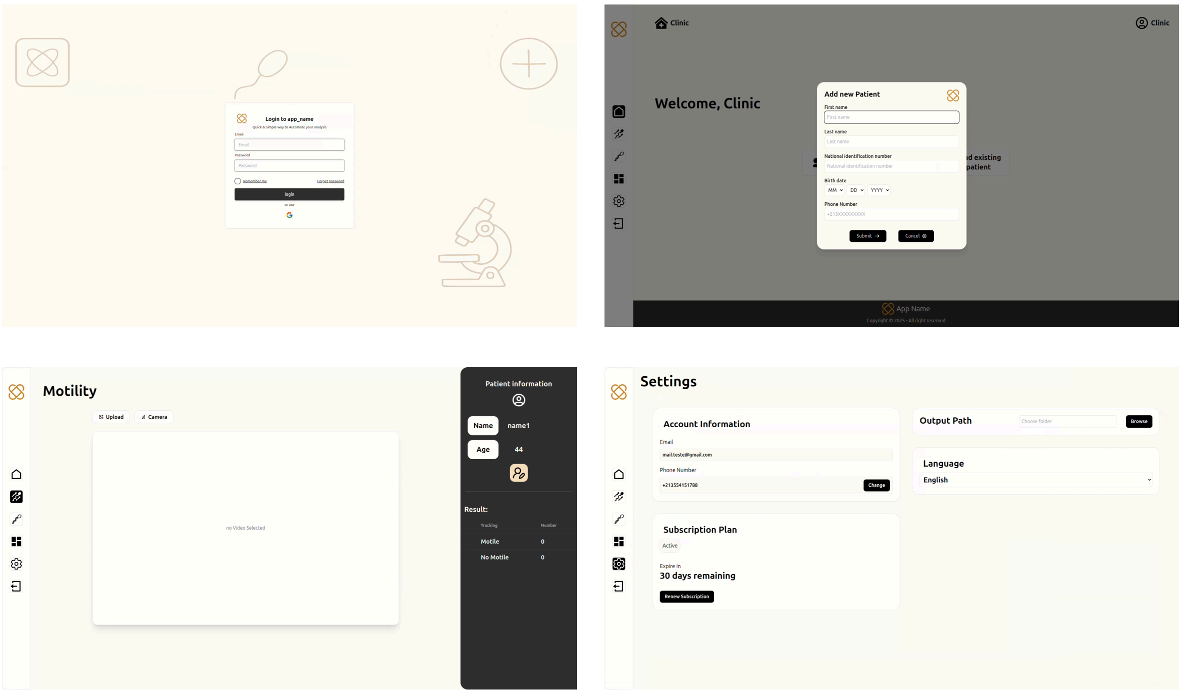


Figure 36 : UI/UX mockup of our application

6.4 Future Deployment Architecture and System Design

As the system transitions from development to production, careful planning of its deployment architecture is essential to ensure scalability, accessibility, and maintainability across various environments. The envisioned deployment strategy follows a hybrid model, combining local processing capabilities with cloud-based infrastructure to balance speed, flexibility, and user accessibility.

The core components of the deployment architecture are outlined below:

Desktop Client (Local Analysis Interface):

The desktop client acts as the central hub for laboratory technicians and veterinarians, offering a local interface tailored to real-time sperm analysis. Built for seamless integration with microscope cameras, it enables direct acquisition and processing of video streams. The embedded AI models perform detection, tracking, and classification tasks on the fly, ensuring immediate feedback for clinical decisions. Designed with reliability in mind, the desktop application also allows reports and media to be cached locally, supporting offline operation in environments with limited internet connectivity.

Cloud Server (Model Updates & Centralized Services)

Complementing the local client, the cloud server plays a vital role in central management and continuous improvement of the system. It is responsible for securely distributing updated AI models to clients, ensuring that the analysis engine remains up to date. Beyond version control, the cloud infrastructure manages essential backend operations such as clinic subscriptions, user profiles, and usage analytics. It also synchronizes encrypted reports and user data, offering centralized access and backup, which is especially valuable for multi-branch clinic networks.

Web Dashboard (Administration & Reporting Portal)

To facilitate oversight and administrative control, a web-based dashboard is provided for clinic managers and administrators. This intuitive portal enables easy handling of subscription plans, billing cycles, and user management. Through the dashboard, managers can monitor system usage, track report generation trends, and review feedback submitted by end users. It also grants secure access to historical records and patient analyses, contributing to transparent and well-documented reproductive health management.

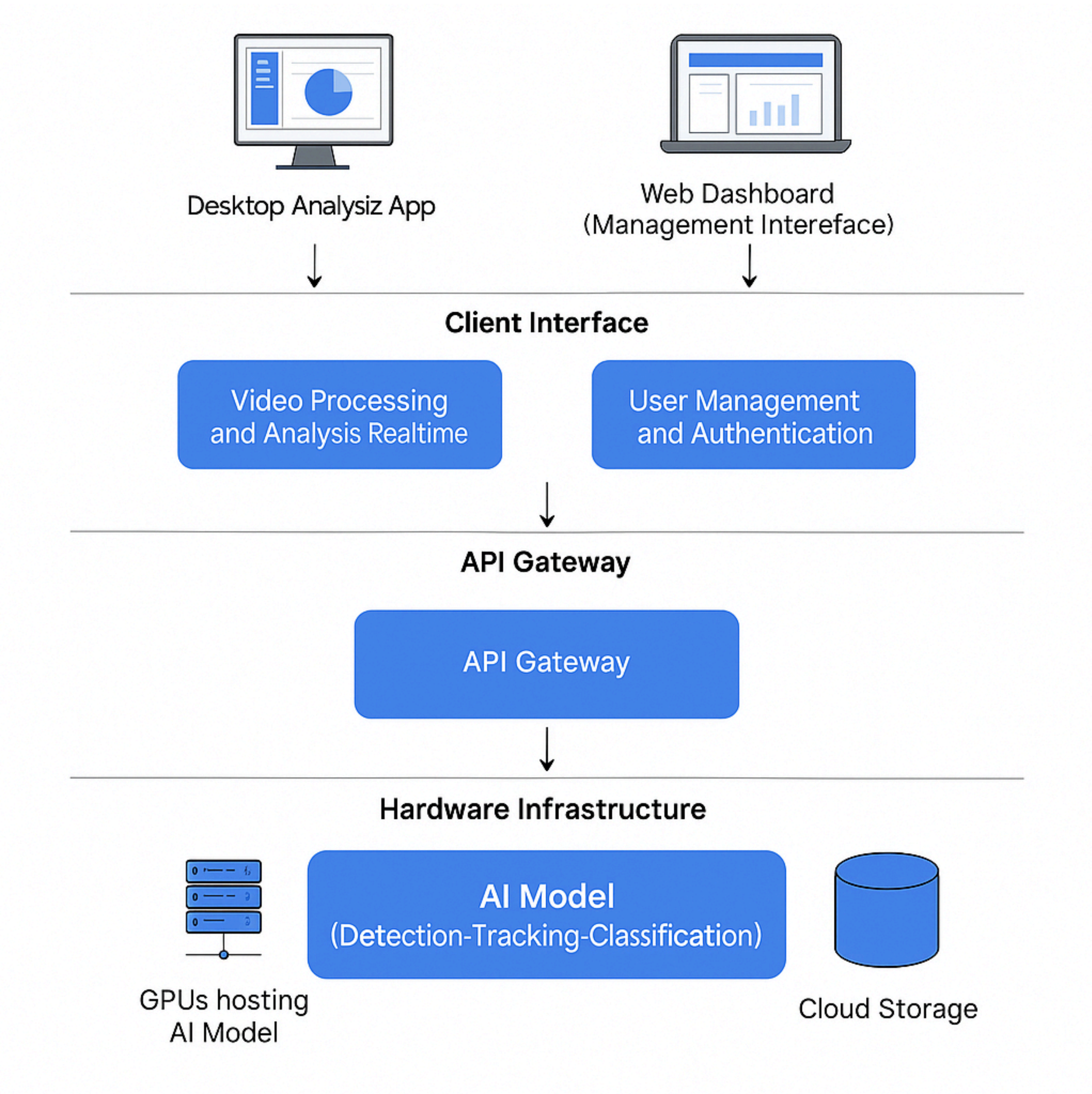


Figure 37:System Architecture Diagram

6.5 Challenges, Ethical Considerations, and Future Work

This chapter explores the key challenges and ethical considerations surrounding AI-driven semen analysis, while also outlining directions for future research. It begins by addressing issues like cross-species generalization and the difficulties posed by limited or imbalanced datasets. The chapter then highlights important ethical aspects of applying AI in reproductive science. Finally, it discusses opportunities to broaden the scope of analysis by including more

species and emphasizes the need for explainable AI to increase transparency and trust in sperm classification models.

6.5.1 Cross-Species Generalization and Domain Adaptation

One of the major challenges in AI-based semen analysis is ensuring robust model generalization across species. Variability in sperm morphology, motility patterns, and image quality between species (e.g., human, bovine, ovine, and rodent sperm) can cause domain shift problems, leading to a drop in performance when models trained on one species are applied to another. Domain adaptation techniques such as feature normalization, adversarial learning, and domain-specific augmentation are increasingly being explored to bridge this gap [58].

6.5.2 Limitations of Current Datasets and Data Imbalance

Most available sperm datasets are small in size and often skewed toward specific species or sperm morphologies. This data imbalance can cause biased learning and poor generalization, particularly when rare species or abnormal cases are underrepresented. Addressing this requires careful dataset curation, balancing techniques such as oversampling or class weighting, and synthetic data generation methods [59].

6.5.3 Ethical Use of AI in Reproductive Science

The application of AI in reproductive technologies raises ethical considerations including data privacy, transparency of algorithms, and the potential for over-reliance on automated systems in critical clinical or veterinary decisions. While automation can assist, it must not replace expert judgment. Equitable access and responsible use in both human and animal fertility applications are central to ethical deployment [60].

6.5.4 Expanding the Dataset to More Species

To ensure the robustness and generalizability of AI models for semen analysis, expanding the dataset to encompass a variety of animal species is a necessary step. Sperm morphology and motility can vary widely between species, influenced by factors such as reproductive strategies, physiology, and evolutionary pressures. Thus, a model trained exclusively on one species may not perform adequately on others unless it is exposed to this variability during training.

By incorporating data from domestic animals such as cattle, sheep, goats, and horses, which are commonly used in breeding programs, the system can support broader veterinary applications. Moreover, wildlife conservation efforts could benefit from the inclusion of endangered species, where semen analysis is critical for assisted reproduction programs.

Open-access datasets like those mentioned in the SpermTree project can guide the structuring of future collections and ensure proper morphological representation across taxa. Even if image data is not always available, taxonomy-aligned metadata (e.g., morphological traits, motility parameters) helps identify the variability that AI systems must account for when scaling [61].

6.5.5 Toward Explainable AI for Sperm Classification

The integration of explainable AI (XAI) is essential to build trust in automated semen analysis systems. Techniques such as Grad-CAM or attention maps help visualize decision rationale, enabling researchers and clinicians to interpret misclassifications or borderline cases. XAI also aids in regulatory acceptance and ensures safer deployment in clinical and breeding programs [62].

6.6 Conclusion

This chapter addressed both the practical deployment and developmental challenges of an AI-powered semen analysis system. On the deployment side, it emphasized integration into existing laboratory workflows, the creation of intuitive clinical interfaces, and a scalable architecture designed to support wide adoption in fertility management and breeding programs. Simultaneously, it explored the core challenges faced during development, such as cross-species generalization, limited and imbalanced data, and the importance of ethical responsibility. Addressing domain shifts, curating diverse datasets, and promoting equitable AI deployment are critical to ensuring robustness and fairness. Future directions include expanding species coverage, incorporating explainable AI, and aligning system transparency with ethical standards to maximize clinical and veterinary impact.

General Conclusion

This engineering project set out to explore and implement an intelligent system for automated sperm analysis using artificial intelligence. At its core, the project aimed to demonstrate the feasibility of using computer vision and deep learning models to perform accurate detection, tracking, and classification of sperm cells under microscopy primarily for veterinary and reproductive science applications.

A major focus was placed on data preprocessing, which played a critical role in shaping the overall performance of our models. Given the sensitivity and complexity of biological imaging, particular care was taken in building a dataset that was both clean and meaningful. We manually curated and annotated microscopy videos, filtered out unusable frames, and balanced sample distributions as much as possible. Advanced preprocessing techniques were applied such as frame normalization, contrast enhancement, and data augmentation to simulate diverse conditions and increase model generalization. The quality of this preprocessing pipeline directly influenced the accuracy and robustness of the AI models developed.

On the machine learning side, we employed state-of-the-art object detection and classification models, with a particular emphasis on the YOLOv11 architecture. This choice was motivated by its balance of speed and accuracy, critical for real-time video analysis. The object detection model was trained to identify individual sperm cells with high precision, while separate classification components were designed to assess their motility and morphology. The pipeline was rigorously evaluated using metrics such as precision, recall, F1-score, and confusion matrices to ensure scientific validity.

Although our primary focus was AI, we also proposed a hybrid software architecture that bridges real-time desktop-based analysis with cloud-based inference and storage. While the full implementation of this software stack remains a future milestone, its conceptual design is fully aligned with the needs of clinical labs and veterinary clinics.

7.1 Limitations and Future Work

While the results obtained throughout this project are promising, several limitations were identified that can guide future improvements.

One of the main challenges encountered was the quality and consistency of the microscopy video data. The raw videos varied significantly depending on the lighting, focus, camera calibration, and the microscope settings used during capture. Such inconsistencies can introduce noise and affect both detection and classification accuracy. Additionally, obtaining high-quality annotated videos required significant manual effort, particularly when aligning camera hardware with microscopes under precise biological conditions. Automating or standardizing this acquisition process could greatly enhance data quality and reduce preprocessing overhead in the future.

Another important limitation is the real-time integration and deployment of our models within a clinical workflow. While the models were validated on preprocessed datasets, they have not yet been tested under continuous, live feed conditions in operational environments. Future work should involve deploying the models within the desktop application and benchmarking their performance in real use cases, particularly focusing on frame rate stability, latency, and feedback speed.

In terms of interpretability, deep learning models still behave as black boxes to a large extent. As the system is intended for medical and veterinary professionals, there is a clear need to incorporate explainable AI techniques. Providing visual justifications for model decisions especially for sperm classification outcomes would not only increase trust but also support more informed clinical decisions.

Moreover, while the project primarily focused on AI development and data processing, further attention should be given to user-centric interface improvements and lab workflow integration in the desktop app and future web dashboard.

Finally**,** this project successfully demonstrated that deep learning techniques, when combined with rigorous data preprocessing and tailored to the specificities of biological imaging, can effectively automate complex reproductive diagnostics. The foundation laid here paves the way for continued research and engineering to build a fully operational and scalable AI system that meets the high standards of accuracy, interpretability, and ethical responsibility required in veterinary reproductive science

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